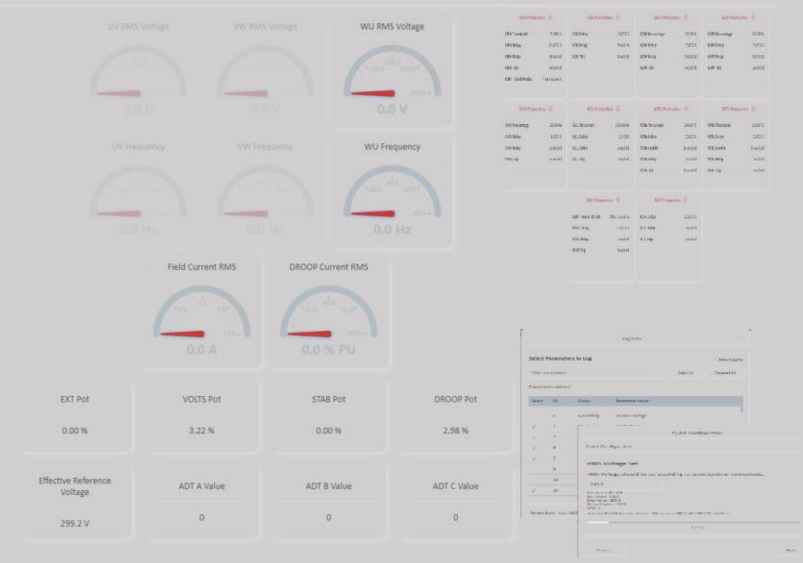


Power Cloud MKII

USER MANUAL

Table of Contents

Revision History

Revision	Revision Date	Explanations	PR
R01	09.10.2025		Ö.K.

1. Introduction

Power Cloud MKII is software developed for monitoring, controlling, and managing ENKO Control Systemsindustrial devices. With its user-friendly interface, powerful features, and flexible structure, it enables you toeasily manage your industrial processes.

This user manual explains all features and usage methods of Power Cloud MKII software in detail. You willlearn all functions step by step, from initial setup to advanced configurations, from basic monitoring to complexdata analysis.

What You'll Find in This Manual:

- **Getting Started and Connection:** Device scanning, login procedures, and initial connection setup
- **Parameter Management:** Parameter reading, writing, import/export operations
- **Data Monitoring:** Real-time graph monitoring, log records, and alarm history
- **Configuration:** Quick configuration, wiring diagram, background image, firmware update
- **Language and Theme:** Multi-language support, language file creation, device language change, themecustomization
- **Advanced Features:** Layout management, offline modes, documentation access

Who Is This For:

This user manual is designed for users with different knowledge levels:

- **Operators:** Daily monitoring and basic control operations
- **Technicians:** Parameter settings, maintenance, and troubleshooting
- **Engineers:** Advanced configuration, system optimization, and analysis
- **System Administrators:** Installation, language file creation, and customization

How to Use This Manual:

- If using for the first time, start by reading all sections in order
- If researching a specific feature, you can go directly to the relevant section
- Each section is supported with step-by-step instructions and visual examples
- Troubleshooting sections provide solutions to common problems you may encounter

Important Information:

Security: Always make backups before critical parameter changes and check device documentation.

Updates: This manual is prepared for the latest version of Power Cloud MKII. New features may be addedwith software updates.

Support: For information not found in the manual or technical issues, you can contact our technical supportteam.

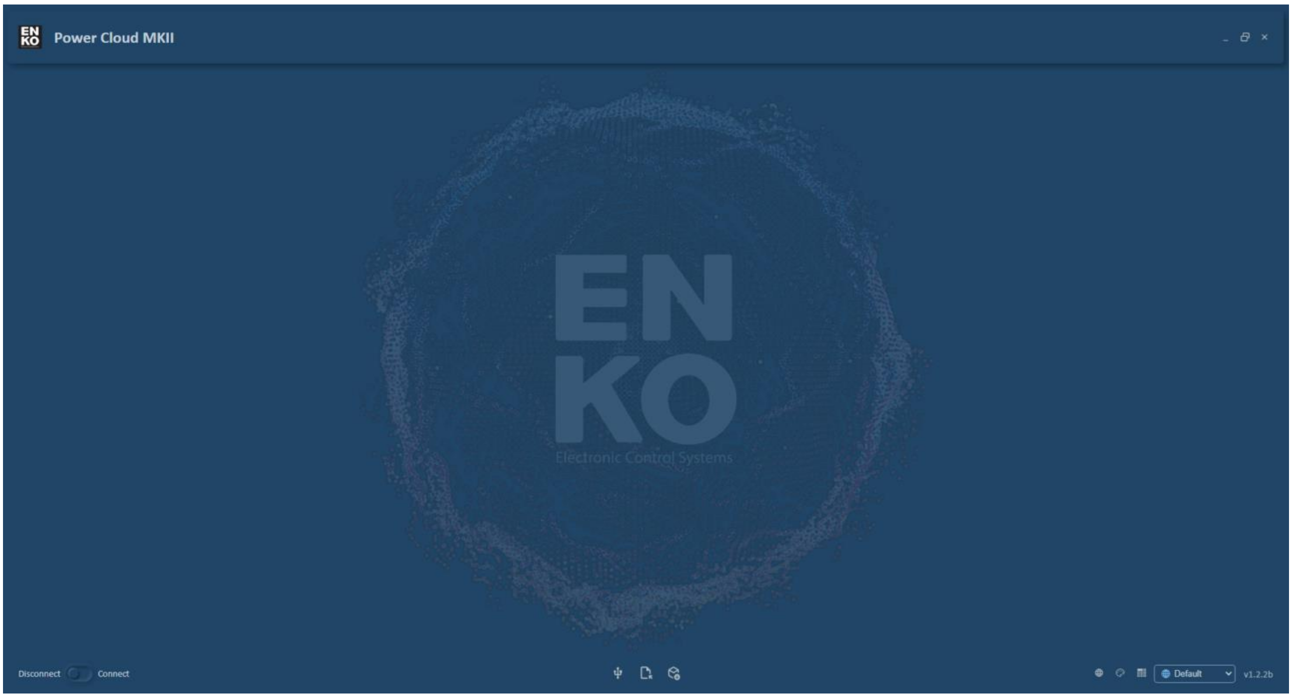
2. Home Page and General Usage

The Power Cloud MKII home page is the starting point of the software and provides access to all basic controls.Thanks to its dynamic JSON-based structure, it offers customized interfaces for each device.

2.1 First Launch

When Power Cloud MKII is started, **a blank page** opens.

First Opening View:



First Launch Features:

- No device connection yet
- Data not visible

2.2 Main Page Button Layout

On the main page, buttons are positioned in three areas:

Group - 1:

Disconnect/Connect Toggle Button

- For establishing or disconnecting device connection
- Toggle button
- **Connect state:** For establishing connection
- **Disconnect state:** For disconnecting connection

Group - 2:

Com. Settings Button

- Opens Communication Settings page
- Configure communication parameters
- **Only visible when not connected**

- Disappears after connection is established

Import File Button (Offline Mode)

- For opening PRL files in offline mode
- View parameters without device connection

Firmware Update Button (Offline Mode)

- Firmware upload without device connection
- For devices in bootloader mode

Group - 3:

Generate Language File Button

- Opens language file creation page
- For creating new language files
- Customize Power Cloud MKII interface language

Theme Button

- Opens Theme Settings page
- Color scheme and object scale settings
- Visual customization

Layout Button

- Opens layout selection menu
- Switch between different screen layouts
- **Used after connecting to device**

Language Selection Box

- For selecting Power Cloud MKII interface language
- Drop-down menu format
- **Options:**
  - Default (English)
  - User Define (User-defined language file)
- Selected before establishing connection

2.3 Device Connection Process

Step 1: Click Connect Button

When the **"Connect"** button is pressed:

Automatic Scanning Starts

- Settings from Com Settings page are used
- Determined protocol (Modbus RTU/TCP) is applied
- All ports or defined IP address are scanned
- Compatible devices are searched

Scan Duration

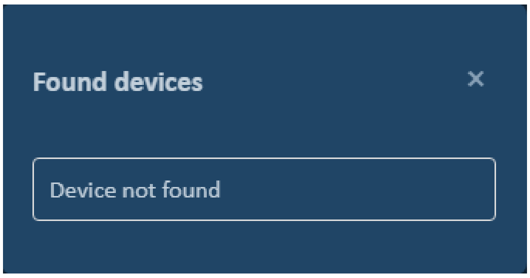
- Usually takes 2-4 seconds
- "Scanning Devices" is displayed
- "Please wait..." message

Step 2: Scan Result

Two possible situations when scanning is complete:

Situation 1: Device Not Found

If no compatible device is found:

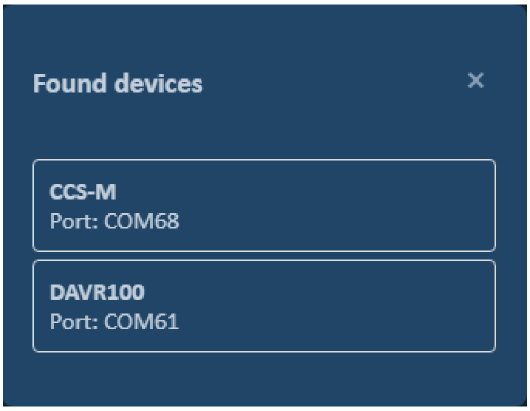


"Device not found" Solution Steps:

- Check physical connections
- Ensure device is powered on
- Review Com Settings
- Verify parameters like Slave Id, IP address
- Try different port
- Check cables

Situation 2: Device Found - Device List

If one or more devices are found, the **"Found Devices"** list opens:



List Content:

- **Device Name:** Device model name

- **Port Information:**
  - For Modbus RTU: COM port (e.g., COM3, COM5)
  - For Modbus TCP: Port (e.g., 502)

Step 3: Device Selection

- Click on the device you want to connect from the list
- Device list **automatically closes**
- **Login screen** opens

Step 4: Login Process

When login screen opens:

- Enter password
- Press **Login** button
- Password is verified

Step 5: Successful Connection

After correct login:

- Login screen closes
- **Main page opens**
- **Device data appears**
- Real-time data flow begins
- Disconnect/Connect button remains in **"Connect"** state
- Com Settings button **disappears**
- Device-specific page layout is loaded

2.4 JSON-Based Dynamic Structure

Power Cloud MKII has a **JSON-based** structure. This means the screen design is completely **dynamic and customizable**.

Advantages of JSON-Based Structure:

Device-Specific Interfaces

- Different screen layout for each device model
- Optimized view according to device features
- Automatic loading and configuration

Dynamic Content

- Screen elements are read from JSON file
- Objects, graphics, buttons are dynamically created
- Flexible and expandable structure

Easy Update

- Just add JSON file for new device support
- No code changes required
- Fast deployment

Multiple Page Support:

Power Cloud MKII can have **multiple monitoring and parameter pages**:

- **Main Dashboard:** General status and basic controls
- **Monitoring 1:** First monitoring page
- **Monitoring 2:** Second monitoring page
- **Parameters:** Parameter lists
- **Advanced:** Advanced settings
- **Etc...**

Device-Specific Differences:

Different screens can come for each device:

- **Device A:** 3 pages (Dashboard, Parameters, Logs)
- **Device B:** 5 pages (Dashboard, Monitoring, Parameters, Maintenance)
- **Device C:** 2 pages (Simple View, Advanced View)

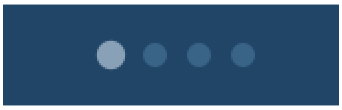
Note: The number of pages and what is shown on each page depends entirely on the device's JSON configuration file.

2.5 Page Navigation

When there are multiple pages, navigation controls are available to switch between pages.

Page Indicator Dots

In the **center** (bottom section) of the main page, **dot indicators** are located.



Dot Indicators:

Dot Features:

- **Total number of dots = Total number of pages**
- **Filled dot (●):** Currently displayed page
- **Empty dot (○):** Other pages
- **Clickable:** Click on dot to go to that page

Navigation with Dots:

Step 1: See Page Count

- Check dots at bottom
- Total number of dots = how many pages there are

Step 2: Go to Desired Page

- Click on the **dot** of the page you want to go to
- Page **slides** to that side
- New page is displayed
- Relevant dot becomes filled (●)

Keyboard Navigation

You can also navigate pages with keyboard arrow keys:

Right Arrow (→):

- Goes to next page
- Page slides to the right
- No next if on last page (no effect)

Left Arrow (←):

- Goes to previous page
- Page slides to the left
- No previous if on first page (no effect)

Down Arrow (↓):

- Goes directly to **last page**
- For quick access
- Jumps to the very end regardless of page count

Up Arrow (↑):

- Goes directly to **first page**
- For quick access
- Always returns to page 1

2.6 Main Page View After Connection

After successfully connecting to the device, the main page is **filled with device-specific content** and additional function buttons become visible.

Connected State Main Page Views:

Dynamic Buttons After Connection

After connecting to the device, additional function buttons become visible **according to device features**. These buttons may vary from device to device.

Dynamic Button Features:

- Features supported by each device may differ
- Some buttons only appear on supporting devices
- Number and arrangement of buttons is device-specific

Button List After Connection

Function buttons that may appear after connection:

Import File

- Uploads parameter files (PRL) to device
- Restores previously exported configurations

Export File

- Saves device parameters to file
- Export in Excel or PRL format

Log Data

- Periodic recording of selected parameters
- Create log file in CSV or TXT format

Update Device Time

- Appears on devices with RTC (Real-Time Clock) feature
- Sets computer time to device
- Timestamp synchronization

Quick Config

- Quick configuration wizard
- Step-by-step basic parameter setup

Wiring Diagram

- Displays device connection diagrams
- Socket pin assignments and cable connections

Chart

- Real-time graph display
- Monitor 4 parameters simultaneously

Alarm History

- Display past alarm records
- Filtering and export features

Event History

- Display system event records
- Same structure as Alarm History

Background Image

- Customize device screen startup image
- Background color, progress color, and QR code content settings

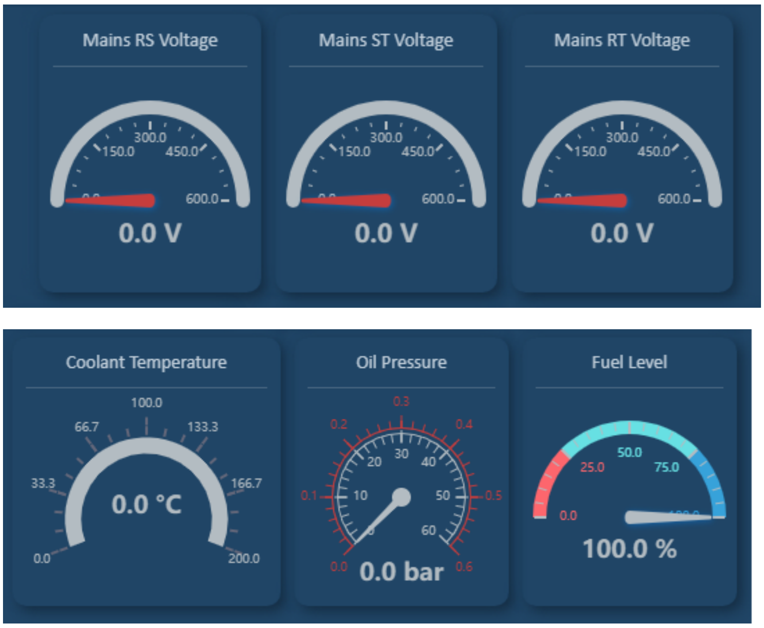
Firmware Update

- Update device software
- Upload firmware in BLF format

Used Objects

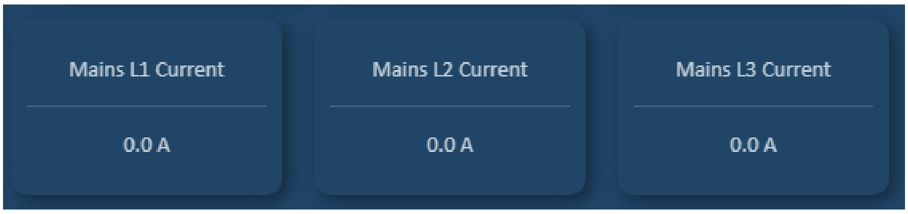
On Power Cloud MKII pages, various objects are used to visualize and control device data. These objects are dynamically created according to JSON configuration.

Gauge (Indicator)



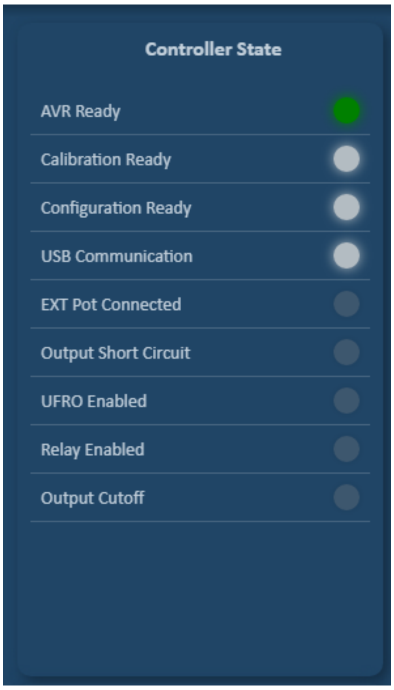
Visualization of data such as voltage, power, frequency, pressure, temperature with circular gauge. When clicked, Parameters Settings page opens and detailed information about the parameter is obtained.

Data Cards



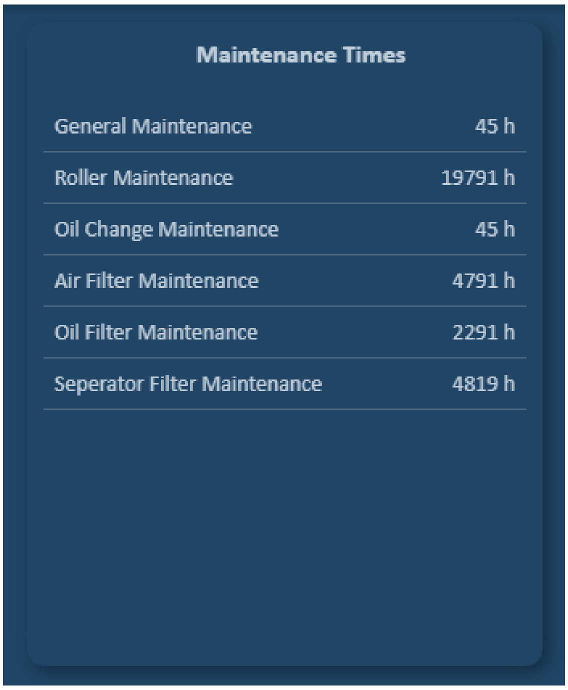
Compact information display in card format where all data can be viewed. When clicked, Parameters Settings page opens and value can be changed.

LED Indicator Lists



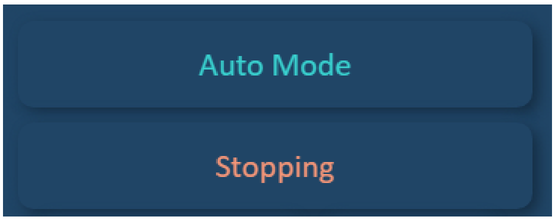
LED indicators that flash according to status information (Input/Output states, Active alarms, etc.).

Data Lists



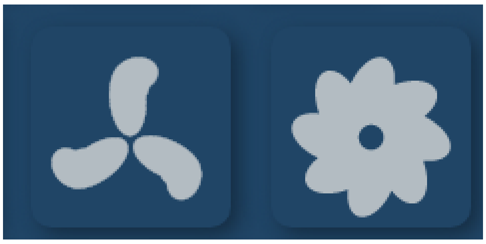
Sequential listing of several parameter information appearing as a list. Used for quick overview.

Status Messages



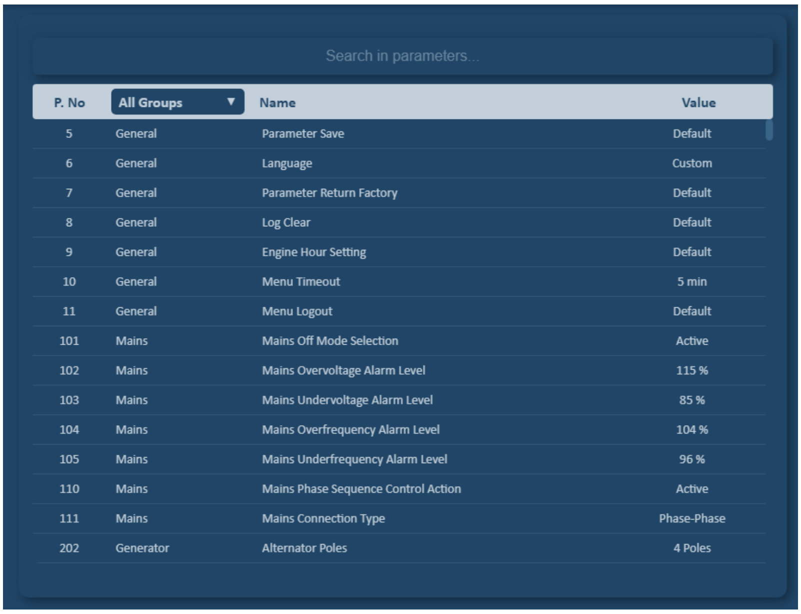
Dynamic message areas composed of texts. Device status information can be displayed, used for group titles and description messages. Text can change according to status.

Image Objects



Dynamic or static image display. Images can change according to status (e.g., different visuals according to operating status).

Table



P. No	All Groups	Name	Value
5	General	Parameter Save	Default
6	General	Language	Custom
7	General	Parameter Return Factory	Default
8	General	Log Clear	Default
9	General	Engine Hour Setting	Default
10	General	Menu Timeout	5 min
11	General	Menu Logout	Default
101	Mains	Mains Off Mode Selection	Active
102	Mains	Mains Overvoltage Alarm Level	115 %
103	Mains	Mains Undervoltage Alarm Level	85 %
104	Mains	Mains Overfrequency Alarm Level	104 %
105	Mains	Mains Underfrequency Alarm Level	96 %
110	Mains	Mains Phase Sequence Control Action	Active
111	Mains	Mains Connection Type	Phase-Phase
202	Generator	Alternator Poles	4 Poles

Displays specific groups in table format. Columns: **P.No, Group, Name, Value**. When clicked, the Parameters Settings page opens for the relevant parameter and the value can be changed.

Table Features:

- **Group Selection Box:** Specific groups can be listed and selected in table header section
- **"Search in parameters..." Entry Box:** Parameter search can be done at the top
- **Search:** Search can be done for columns other than Value (P.No, Group, Name)

2.7 Disconnect Process

To disconnect:

Step 1: Click Disconnect Button

- Press the **"Disconnect"** button in the lower left corner

Step 2: Connection Terminates

- Device connection is terminated
- Data flow stops
- Values disappear

Step 3: Page Returns to Initial State

- Main page becomes blank page again
- Com Settings button reappears
- Layout button remains disabled
- Disconnect/Connect button is in "Disconnect" state

2.8 Main Page Summary

First Launch:

- Blank page view
- Connection established with Connect button
- Com Settings, Theme, Language controls ready

Connection Process:

1. Connect → Scanning
2. Device list or "Device not found"
3. Device selection
4. Login
5. Main page data loads

Navigation:

- □ Page viewing with dot indicators
- ⌨Quick switch with keyboard arrow keys
- Click and drag support

Dynamic Structure:

- □ JSON-based configuration
- Device-specific interfaces
- □ Multiple page support
- □ Customizable content

3. Communication Settings Page

The Communication Settings page allows you to configure communication parameters between Power Cloud MKII and the device. This page is used to determine the communication protocol and related settings before connecting to the device.

3.1 Opening Communication Settings Page

Com. Settings Butonunun Görünürlüğü:

The visibility of the Communication Settings button depends on device connection status:

BEFORE Connection:

- **Com. Settings** button is **visible** and active
- You can open Communication Settings page by clicking the button
- You can configure connection parameters

AFTER Connection:

- **Com. Settings** button **disappears** (becomes invisible)
- Communication settings cannot be changed during active connection
- To change settings, you must first **Disconnect**

3.2 Communication Settings

Communication Protocol Selection

The first and most important option on the Communication Settings page is **Communication** (Communication Protocol) selection.

Communication Selection Box:

Two main protocol options are available:

- **Modbus RTU** - Serial communication protocol
- **Modbus TCP** - Ethernet/IP-based communication protocol

Depending on the protocol you select, the page content **changes dynamically** and relevant parameters are shown.

Modbus RTU Settings

When **Modbus RTU** is selected, the following parameters for serial communication are displayed:

Modbus RTU Parameters:

1. Slave ID

- Device identification number on Modbus network
- Default: Usually 1
- Example: 1, 2, 10, 100

2. Communication Timeout

- Communication timeout duration
- If no response within this time, connection is considered faulty

- **Unit:** Milliseconds (ms)
- **Recommended Value:** 1000 - 2000 ms
- **Example:** 1000 ms (1 second)

3. Refresh Rate

- How often data will be updated
- Faster value = More frequent updates
- **Unit:** Milliseconds (ms)
- **Recommended Value:** 50 - 1000 ms
- **Example:** 500 ms (2 updates per second)

4. Baud Rate

- Serial communication speed
- Data transmission rate in bits/second
- **Standard Values:**
  - 9600 bps
  - 19200 bps
  - 38400 bps
  - 57600 bps
  - 115200 bps
- **Default:** 115200

5. Data Bits

- Number of bits in each data packet
- **Options:**
  - 7 bits
  - 8 bits
- **Default:** 8 bits

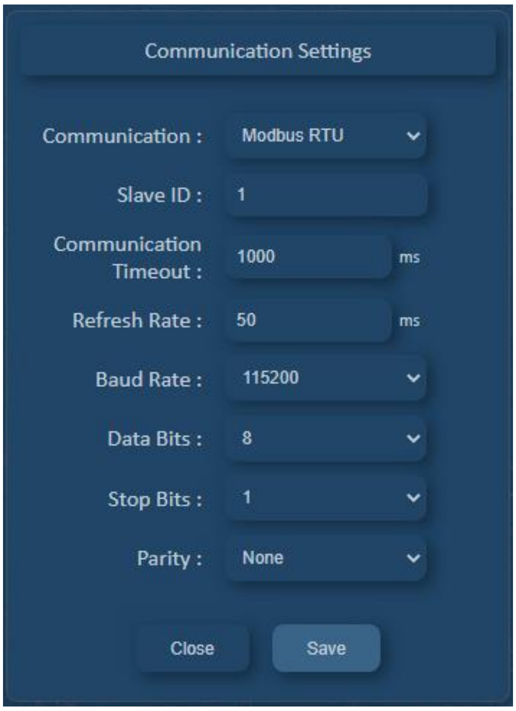
6. Stop Bits

- Stop bits at end of data packet
- **Options:**
  - 1 bit
  - 2 bits
- **Default:** 1 bit

7. Parity

- Parity bit used for error checking
- **Options :**
  - **None:** No parity
  - **Even:** Even parity
  - **Odd:** Odd parity
- **Default:** None

Modbus RTU Page View:



Modbus TCP Settings

When **Modbus TCP** is selected, page content changes and parameters required for Ethernet/IP communication are shown.

Modbus TCP Parameters:

1. IP Address

- Device's IP address on the network
- **Format:** xxx.xxx.xxx.xxx
- **Example:**
  - 192.168.1.100
  - 10.0.0.50
  - 172.16.0.200

2. Slave ID

- Device ID also used in Modbus TCP
- May not be important on some devices (usually 1 is used)
- **Example:** 1

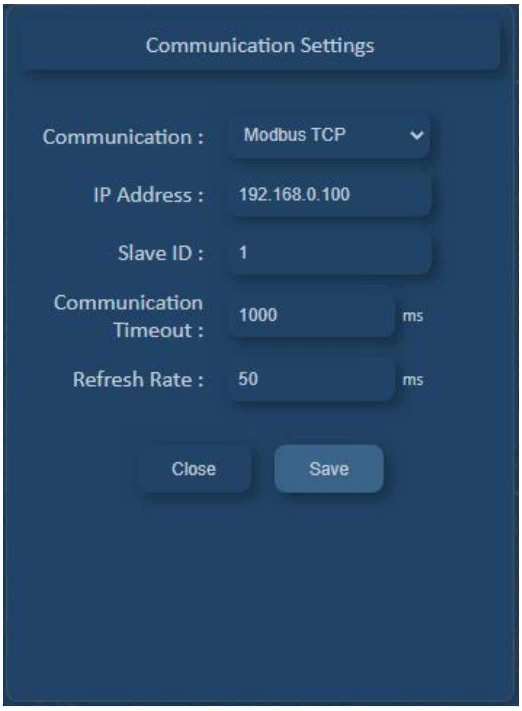
3. Communication Timeout

- Communication timeout duration
- **Unit:** Milliseconds (ms)
- **Recommended Value:** 1000 - 2000 ms
- **Example:** 1000 ms (1 second)

4. Refresh Rate

- How often data will be updated
- **Unit:** Milliseconds (ms)
- **Recommended Value:** 50 - 1000 ms
- **Example:** 500 ms

Modbus TCP Page View:



Note: When Modbus TCP is selected, Baud Rate, Data Bits, Stop Bits, and Parity parameters disappear. These parameters are only valid for serial communication (Modbus RTU).

3.3 Saving Communication Settings

Two buttons are on the Communication Settings page: **Save** and **Close**.

Save Button:

Used to save the communication settings you configured.

Save Process Steps:

Step 1: Set All Parameters

- Select communication type (Modbus RTU or TCP)
- Enter all relevant parameters correctly

Step 2: Press Save Button

- Click the **"Save"** button at the bottom of the page
- Settings are automatically saved

Step 3: Saved Settings Become Active

- Saved settings are now active
- These settings are used when you press **Connect** button
- Automatically loaded in next session

Auto-Load Feature:

- Saved settings are stored **permanently**
- Settings are preserved even if you close Power Cloud MKII
- **Automatically** loaded on next launch r
- Same settings used until manually changed

Close Button:

Used to close the page **without saving** changes you made.

3.4 Settings Change Process

To Change Current Settings:

Step 1: Disconnect Current Connection

- If connected to device, press **Disconnect** button
- Com. Settings button becomes visible

Step 2: Open Com. Settings Page

- Click **Com. Settings** button
- Current settings appear on page

Step 3: Make Changes

- Change desired parameters
- Enter new values

Step 4: Save and Connect

- Press **Save** button
- Return to main screen
- Press **Connect** button to connect with new settings

4. Login Screen

The login screen enables you to establish secure connection between Power Cloud MKII and the device and provides authorization-based access control. Different functions are accessible according to user level.

4.1 Device Scanning and Selection

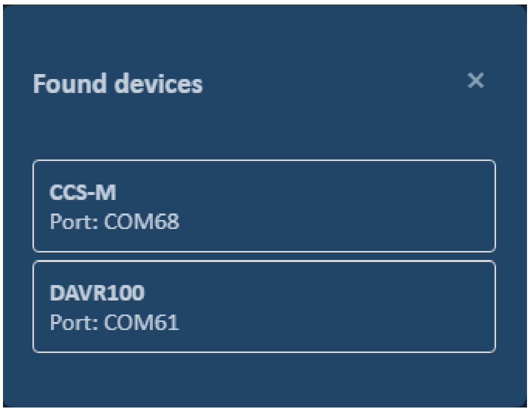
The login process begins with the device scanning step.

Step 1: Click Connect Button

- Press the **"Connect"** button on main page
- Software starts automatic scanning

Step 2: Automatic Scanning

- Software scans according to settings on **Communication Settings** page
- Determined protocol (Modbus RTU/TCP) is used
- Set port and connection parameters are applied
- All compatible devices connected to computer are searched



Step 3: Found Devices List

After scanning completes, a list titled **"Found Devices"** appears on screen.

List Features:

- List of all devices **plugged into the computer**
- **Multiple devices** may be connected to one computer
- Identifying information shown for each device

Step 4: Device Selection

- Click on the device you want to connect from the list
- List automatically **closes**
- **Login screen** opens for selected device

Note: If no device is found, "Device not found" message is displayed. In this case, check Communication Settings and physical connections.

4.2 Login Screen Options

There are **two different login screens** in Power Cloud MKII. Which screen opens **varies by device type**.

Type 1: Login Screen

The first type login screen requires only password entry and user level is automatically determined.

Screen Components:

1. User Authentication Header

- "User Authentication" text at top of screen
- Indicates secure login process

2. Password Entry Box

- Text box for password entry
- Characters appear **hidden** (●●●●●)

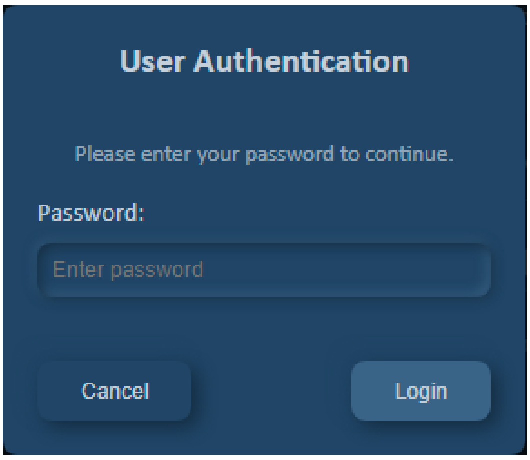
3. Login Button

- Clicked after password is entered
- Password is verified and login is performed

4. Cancel Button

- Cancels login process
- Screen closes and returns to beginning

Screen View:



Type 2: Login Screen

The second type login screen requires the user to first select access level, then enter the password for that level.

Screen Components:

1. User Authentication Header

- "User Authentication" text at top of screen

2. Access Level Selection Box

- Drop-down menu to select user level
- **Options:**
    - **Operator:** Basic usage and monitoring
    - **Service:** Advanced settings and maintenance
    - **Factory:** Full access and factory settings

3. Password Entry Box

- Password for selected level is entered
- Characters appear hidden (●●●●●)

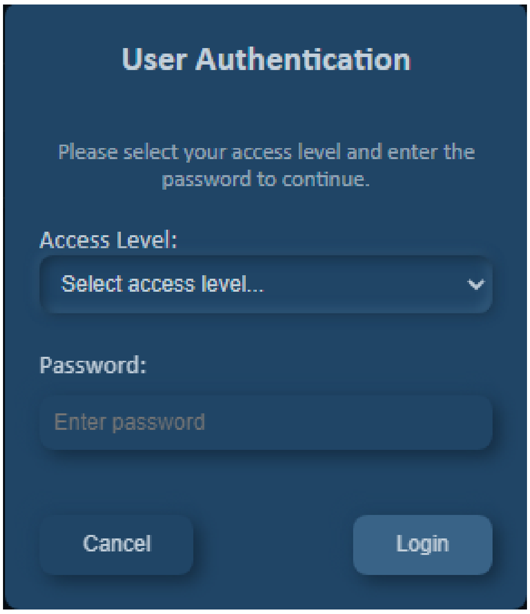
4. Login Button

- Clicked after level is selected and password is entered

5. Cancel Button

- Cancels login process

Screen View:



4.3 After Login

Successful Login:

When login is successful:

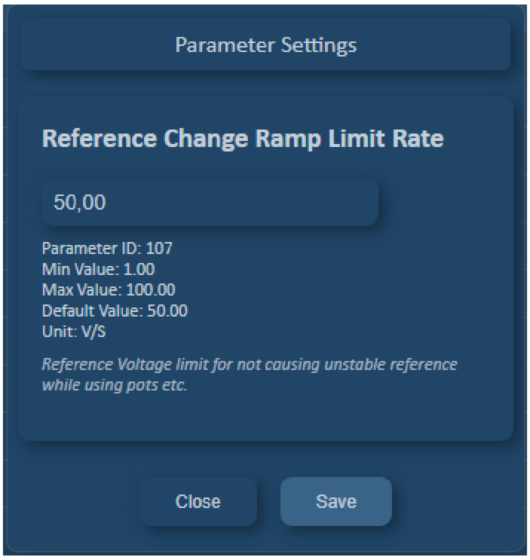
- **Login screen closes**
- **Main screen** opens and device data is displayed
- **User level indicator** appears on screen (lower right corner)
- **Real-time data flow** starts

Level Change:

To switch to different level:

- Press **Disconnect** button to disconnect
- Press **Connect** button again
- Select same device
- Enter **different level/password** on login screen

5. Parameter Settings Page



The Parameter Settings page allows you to view and modify your device's operating parameters. Through this page, you can access detailed information about each parameter and make necessary changes safely.

5.1 Parameter Information

The following information is displayed for each parameter:

- **Current Value:** Parameter's current value on device
- **Minimum Value:** Smallest value parameter can have
- **Maximum Value:** Largest value parameter can have r
- **Default Value:** Parameter's factory default value
- **Unit:** Parameter's unit of measurement (e.g., V, A, Hz, °C)
- **Description:** Brief informational text about parameter

Note: By carefully reviewing parameter information, you can evaluate the effects of changes on device operation beforehand.

5.2 Changing Parameter Value

Numeric Parameters:

If parameter value is numeric:

- Enter new value in **entry box** on screen
- Ensure entered value is between Min and Max values
- Press **Save** key to confirm value

Selection-Based Parameters:

If parameter value contains predefined options:

- Open the **selection box** shown on screen
- Select appropriate option (e.g., Active/Passive, On/Off)
- Press **Save** key to confirm selection

5.3 Parameter Dependency Control

Power Cloud MKII automatically checks relationships between parameters and prevents possible errors.

Dependency Control Features:

- If a parameter's value is linked with another parameter, software monitors this relationship
- **Example:** If parameter 100 has condition that it cannot be greater than parameter 101:
  - If new entered value violates this rule, system gives warning
  - Invalid value not sent to device
  - Error message shown to user

Security Measures:

- All parameter changes verified before sending to device
- Values exceeding Min/Max limits not accepted
- Changes violating dependency rules blocked

5.4 Saving or Canceling Parameter Change

Two buttons on page:

Save Button:

- Enter or select new parameter value
- Press **Save** button
- All checks (Min/Max limits, dependency rules) successfully completed
- Value automatically written to device
- Current value field updated to show new value
- Confirmation message displayed when change successfully saved
- Page automatically closes

Close Button:

- You can press **Close** button at any time
- Page closes
- Changes made **not sent to device**
- Previous value of parameter preserved
- No changes saved

Warning: When you press Close key, all changes you made will be lost. If you want to send changes to device, be sure to press **Save** button.

6. Exporting Parameter Information

Power Cloud MKII allows you to export all parameter values from your device in two different file formats. This feature is used for parameter backup, documentation, and quick configuration transfer.

6.1 Export File Formats

Power Cloud MKII supports two different export formats:

- **Excel Format:** Human-readable, for documentation and reporting
- **PRL Format:** Software-readable, for backup and restore

Each format serves different purposes and you can choose appropriate format according to your needs.

6.2 Parameter Export Process

Export Process Steps:

Step 1: Click Export File Button

- Press **Export File** button from main page
- File save window opens

Step 2: Select Save Location

- Select **folder** where file will be saved in opened window
- Specify **file name**
- You can customize file name if desired

Step 3: Select Save Type

- Select desired format from **Save Type** option:
  - **Excel (.xlsx)**
  - **PRL (.prl)**

Step 4: Press Save Button

- Click **Export** button after making all selections
- File creation process starts
- Confirmation message displayed when complete
- File saved to location you specified

Excel Format

Excel format presents parameter information in human-readable and editable table.

Excel File Content:

File contains table with three columns:

Column Descriptions:

- **Parameter No:** Parameter's number (ID) within device
- **Parameter Name:** Parameter's descriptive name
- **Current Value:** Parameter's value at export time

Excel File Features:

- **All parameters** on device included in list
- Parameters sorted by number
- File can be opened with Microsoft Excel, Google Sheets, and similar programs
- Filtering, sorting, and search operations possible
- Can be printed and used in reports

Excel Format Usage Areas:

- **Documentation:** To document device configuration
- **Reporting:** To be included in technical reports
- **Comparison:** To compare parameter values at different times
- **Analysis:** To examine and analyze parameter values
- **Archiving:** For long-term record keeping
- **Sharing:** To share with team members or technical support

Note: Excel file is only for viewing and documentation. You cannot edit this file and load it back to device. To write parameters to device, use PRL format.

PRL Format (.prl)

PRL format is Power Cloud MKII's proprietary file format designed for parameter backup and restore.

PRL File Features:

- **Software-Readable:** Can be opened directly by Power Cloud MKII
- **Fast Loading:** Can quickly write parameters back to device
- **Compact Format:** Efficient file size
- **Secure:** Contains data integrity check

Important: PRL files should not be opened or edited with text editors. These files should only be processed by Power Cloud MKII.

PRL Format Usage Areas:

- **Backup:** To backup all parameter values
- **Restore:** To write parameters back to device
- **Configuration Copy:** To transfer same settings to another device
- **Quick Setup:** To load pre-prepared setting sets
- **Version Management:** To save different configuration versions
- **Factory Settings:** To save standard configurations

7. Importing Parameter Information

Power Cloud MKII allows you to open previously saved parameter files in PRL format to view parameters and write them to device. Import operation can be done both when device connection exists (online) and when no connection (offline).

7.1 PRL File Import Process

Import Process Starting Steps:

Step 1: Click Import File Button

- Press **Import File** button from main page
- File selection window opens

Step 2: Select PRL File

- Find the **.prl** file you previously saved in opened file selection window
- Select file

Step 3: Press Import Button

- Press **Import** button after file is selected
- **PRL File Data** page automatically opens

7.2 PRL File Data Page

PRL File Data page displays parameter information in file in detail and allows you to manage writing operations to device.

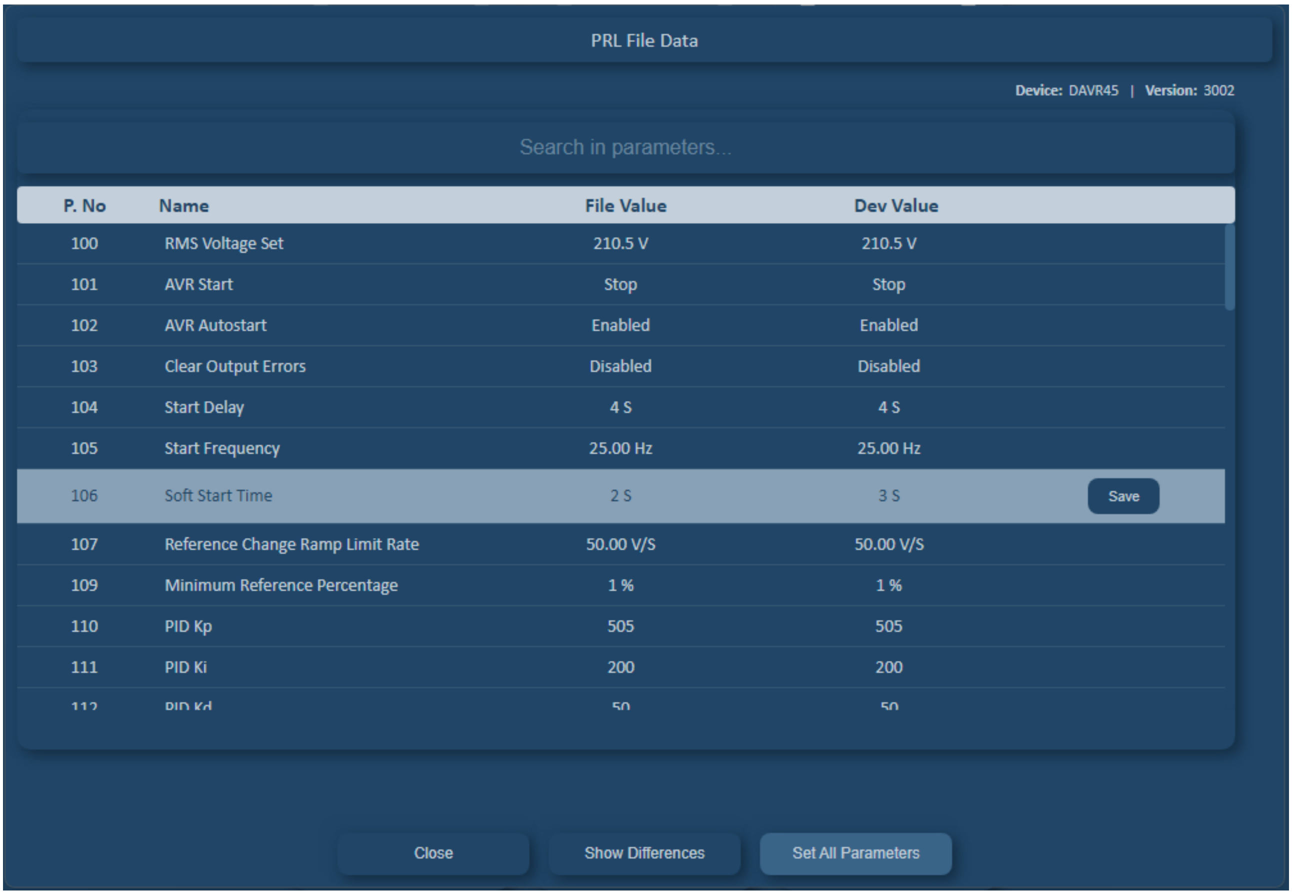
Page Top Section - Device Information:

At top of page, following information about PRL file is displayed:

- **Device Model:** Which device model file was created for
- **Device Version:** Which device version file was created on

Parameter Table:

Main section displays table with four columns:



Column Descriptions:

- **P. No:** Parameter's number (ID) within device
- **Name:** Parameter's descriptive name
- **File Value:** Value saved in PRL file
- **Dev. Value:** Current value in device for that parameter (in online mode)

7.3 Online Mode - When Device Connection Exists

When you import while connected to device, values in file are compared with device's current values.

Difference Display:

- Values in **File Value** and **Dev. Value** columns are compared
- **If two values are different**:
  - Row color changes and becomes **prominent/highlighted**
  - **Save** button appears at end of row
  - This allows user to easily see which parameters are different
- **If two values are same**:
  - Row remains normal color
  - Save button doesn't appear
  - Parameter already up to date

Individual Parameter Update:

To update only specific parameter:

- Find row of different (highlighted) parameter
- Click **Save** button at end of row
- Only that parameter written to device with File Value update
- Dev. Value updates when complete
- Row highlight disappears and returns to normal color

Bulk Parameter Update:

To update all different parameters at once:

- Press **"Set All Parameters"** button at bottom of page
- All different parameters automatically detected
- Parameters written to device in sequence
- Dev. Value column updates when all parameters written
- All row highlights disappear

Warning: When you press Set All Parameters button, all different parameters will be changed according to file values. Carefully check values before starting operation.

Search and Filtering

PRL File Data page offers search and filtering features for easy navigation through long parameter lists.

Parameter Search:

Using **"Search in parameters..."** entry box at top of page:

- Type parameter name or number in search box
- List automatically filtered
- Only parameters matching search criteria shown
- When you clear search box, full list reappears

Example Searches:

- Type "Motor" to find all motor-related parameters
- Type "100" to find parameter 100
- Type "Speed" to filter speed-related parameters

Show Differences:

"Show Differences" button at bottom of page:

- When button clicked, list filtered
- Only parameters where **File Value and Dev. Value** differ shown
- Parameters with same value hidden
- This way you can focus only on parameters that need changing
- Click button again or remove filter to see full list again

Tip: If you opened PRL file with many parameters, first press "Show Differences" button to focus only on different parameters. This speeds up your work and prevents confusion.

7.4 Offline Mode - When No Device Connection

Even when not connected to device, you can open PRL files and view content.

Offline Mode Features:

- **Import File** button can be used without device connection
- When PRL file selected, **PRL File Data** page still opens
- However in this case **Dev. Value** column appears **empty**
- Since no device connection, current device values cannot be read

Limitations in Offline Mode:

- **Save** buttons don't appear (individual parameter writing cannot be done)
- **Set All Parameters** button disabled (bulk writing cannot be done)
- **Show Differences** button doesn't appear (comparison cannot be made)
- Data writing operations to device cannot be done
- Dev. Value column remains empty

Available Features in Offline Mode:

- View PRL file content
- Examine parameter names and numbers
- Read values in File Value column
- Perform parameter search (**Search in parameters...**)
- View device information
- Verify file content

Offline Mode Usage Purposes:

- Check PRL file content
- Verify which device file is for
- View parameter values as reference
- Check if file is corrupted

7.5 Import Process Usage Scenarios

Scenario 1: Restore Backup Settings

- Select PRL file you previously exported
- See different parameters with Show Differences
- Restore all settings with Set All Parameters

Scenario 2: Partial Parameter Update

- Open PRL file
- If you want to change only certain parameters
- Use Save buttons on relevant rows
- Other parameters remain unchanged

Scenario 3: Check File Content (Offline)

- Import File without connecting to device
- Review file content
- Confirm parameters are correct
- Connect to device and apply later

Scenario 4: Copy Settings Between Different Devices

- Export parameters from Device A (PRL)
- Connect to Device B
- Import PRL file
- See differences and apply with Set All Parameters

Scenario 5: Verify Same Parameter

- To check a parameter you suspect
- Import your old PRL file
- Find that parameter with Search
- Compare File Value and Dev. Value

7.6 Import Process Precautions

Security and Compatibility:

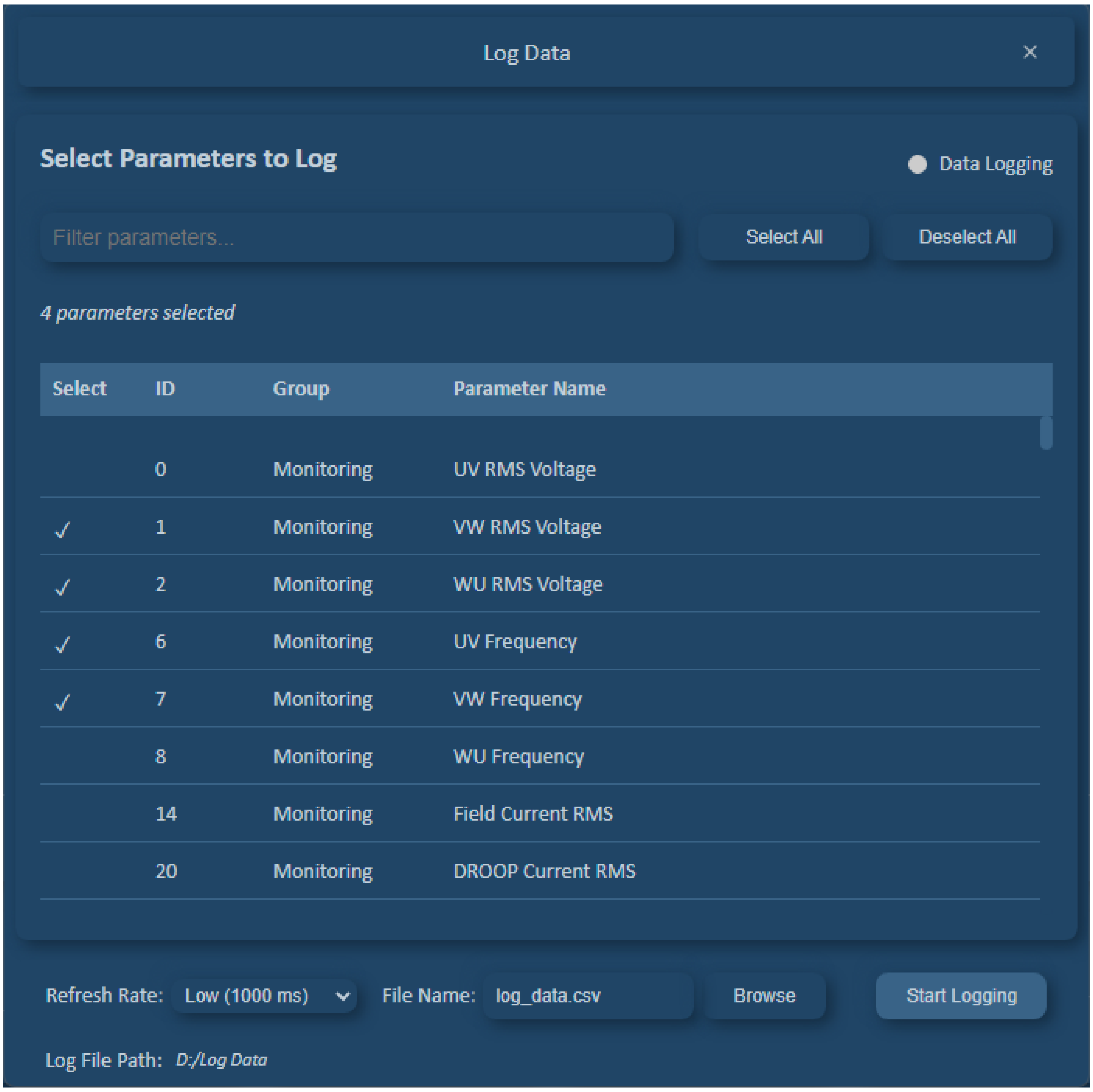
- **Device Compatibility:** Ensure PRL file is for correct device model
- **Firmware Version:** Check file compatible with current firmware version
- **Critical Parameters:** Check before changing security-related parameters
- **Backup:** Export PRL before changing current settings

Error Conditions:

- If file corrupted or unreadable, error message displayed
- If device connection lost, operation stops
- Warning given if incompatible device model detected

Important: Import operation cannot be undone. When you press Set All Parameters or Save buttons, parameters immediately written to device. Therefore, always backup your current settings before operation.

8. Log Data Page



The Log Data page enables automatic recording of data from the device at intervals you specify. On this page, you can configure which parameters will be recorded, how frequently, and in what format.

8.1 Selecting Log Parameters

Parameter List:

The table on screen lists all parameters that can be logged. This list contains device-specific parameters.

Parameter Selection Methods:

- **Manual Selection:**
  - Click on the parameter you want to log in the table
  - Parameter becomes selected
  - You can select as many parameters as you want
- **Filter Search:**
  - Use the **filter box** on screen to search for parameters
  - Select desired parameter from search results
  - Filtering provides quick access in long parameter lists
- **Bulk Selection:**
  - Press **Select All** button to select all parameters

    - Press **Deselect All** button to remove all selections

Note: Selected parameters will be automatically remembered on next connection. Log options are saved separately for each device.

8.2 Log Recording Settings

Refresh Rate (Recording Frequency):

- Select log collection frequency from **Refresh Rate** selection box
- Value entered in milliseconds (ms)

Recommendations:

- Select shorter period for fast-changing data (200-1000 ms)
- Select longer period for slow-changing data (1000-5000 ms)
- Very short periods can cause large file sizes

Recording Location and Format Selection:

- Press **Browse** button
- From opened window:
    - Select **folder** where log file will be saved
    - Specify **file name**
    - Select file format from **save type** option:
        - **CSV format:** Comma-separated values (Excel compatible)
        - **TXT format:** Plain text file
- Confirm settings

8.3 Starting and Stopping Log Recording

Starting Log Recording:

- Select parameters
- Set Refresh Rate value
- Specify recording location and format
- Press **Start Logging** button
- Log process starts and displayed in status bar

Monitoring Log Status:

- When log collection starts, log status appears in **status bar** on main page
- Log process continues in background even if you close Log Data page
- You can switch to other pages and continue other operations
- Status bar continuously shows log process is active

Stopping Log Recording:

- Press **Stop Logging** button
- Log process stops
- Log indicator in status bar disappears
- Saved file becomes ready at location you specified

8.4 Using Log Files

After log files are saved:

- CSV files can be opened with Microsoft Excel or similar programs
- TXT files can be opened with any text editor

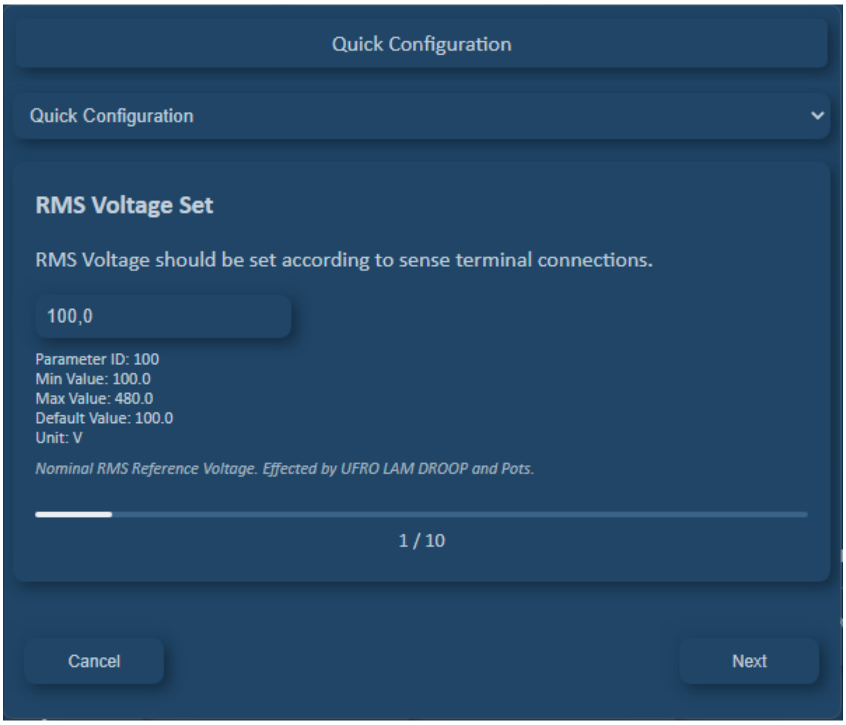
- Data can be used for analysis, chart creation, or reporting
- Files contain parameter values with timestamps

8.5 Important Notes

- **Device-Specific Recording:** Log options saved and remembered separately for each device
- **Auto-Load:** On next connection, parameters you previously selected for that device automatically load
- **Continuous Recording:** Log collection continues working in background
- **File Size:** Long-term logging can create large files, check periodically
- **Connection Loss:** Log process automatically stops if device connection lost

Tip: Check your log settings before critical tests and do a test recording to ensure file format is correct.

9. Quick Configuration Page



The Quick Configuration page allows you to configure basic settings of your device quickly and easily. With step-by-step guidance, you can set parameters correctly and apply changes collectively.

9.1 Quick Configuration Selection

When page opens, **Quick Configuration** selection box is displayed.

Configuration Options:

- Multiple quick configuration profiles may be in selection box
- Each profile contains pre-defined parameter groups for specific application or scenario
- Select configuration profile suitable for your needs
- **Example profiles:** Initial Setup, Standard Settings, Advanced Settings, etc.

9.2 Step-by-Step Parameter Configuration

Quick Configuration provides easy and systematic configuration by showing parameters page by page.

Information on Each Page:

- **Parameter Name:** Name of parameter to be set
- **Current Value:** Parameter's current value
- **Entry Field:** Area where you can enter new value or selection box
- **Description:** Detailed information about parameter and usage purpose
- **Min/Max Values:** Value range allowed for entry
- **Unit:** Parameter's unit of measurement

Parameter Setting Steps:

- Review parameter information displayed on page
- Understand parameter's function by reading description text
- Enter desired value or select from selection box
- If you don't want to change value, you can leave current value as is

9.3 Navigation Between Pages

Next Button:

- Confirms settings on current page
- Moves to next parameter page
- Allows you to proceed if entered value is valid
- Gives warning if invalid value entered

Previous Button:

- Returns to previous parameter page
- You can review changes made on previous pages
- You can rearrange values if necessary
- You can go back as many pages as you want

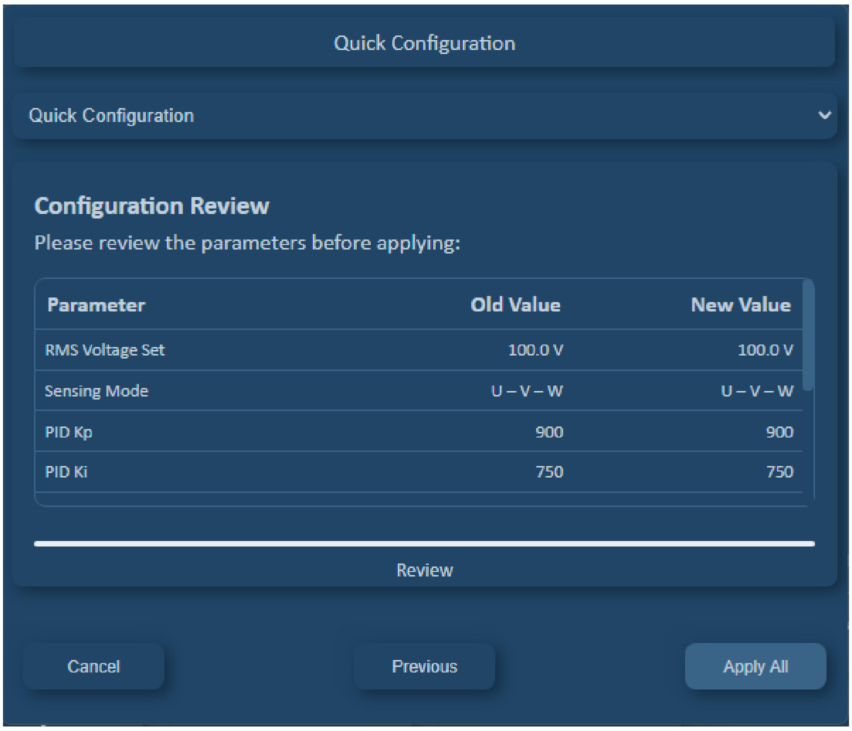
Cancel Button:

- Cancels Quick Configuration process
- All changes made are lost
- No value written to device
- Returns to main screen or previous page

Note: When you press Cancel button, all settings you made so far will be lost. If unsure, proceed to Configuration Review page and see changes before deciding.

9.4 Configuration Review

After completing all parameter pages, **Configuration Review** page is displayed at final step.



Configuration Review Page Content:

All parameters listed in table format on page:

Table Information:

- **Parameter Name:** Name of parameter to be changed
- **Old Value:** Current value on device
- **New Value:** New value to be applied

Checking Changes:

- Carefully review all parameters in table
- Compare old and new values
- If incorrect or unwanted change, you can go back with **Previous** button and correct
- If all changes correct, proceed to next step

9.5 Applying Changes

Apply All Button:

- Press **Apply All** button on Configuration Review page
- All parameters written to device in one operation
- You receive confirmation message when all parameters successfully written
- Page automatically closes and changes become active

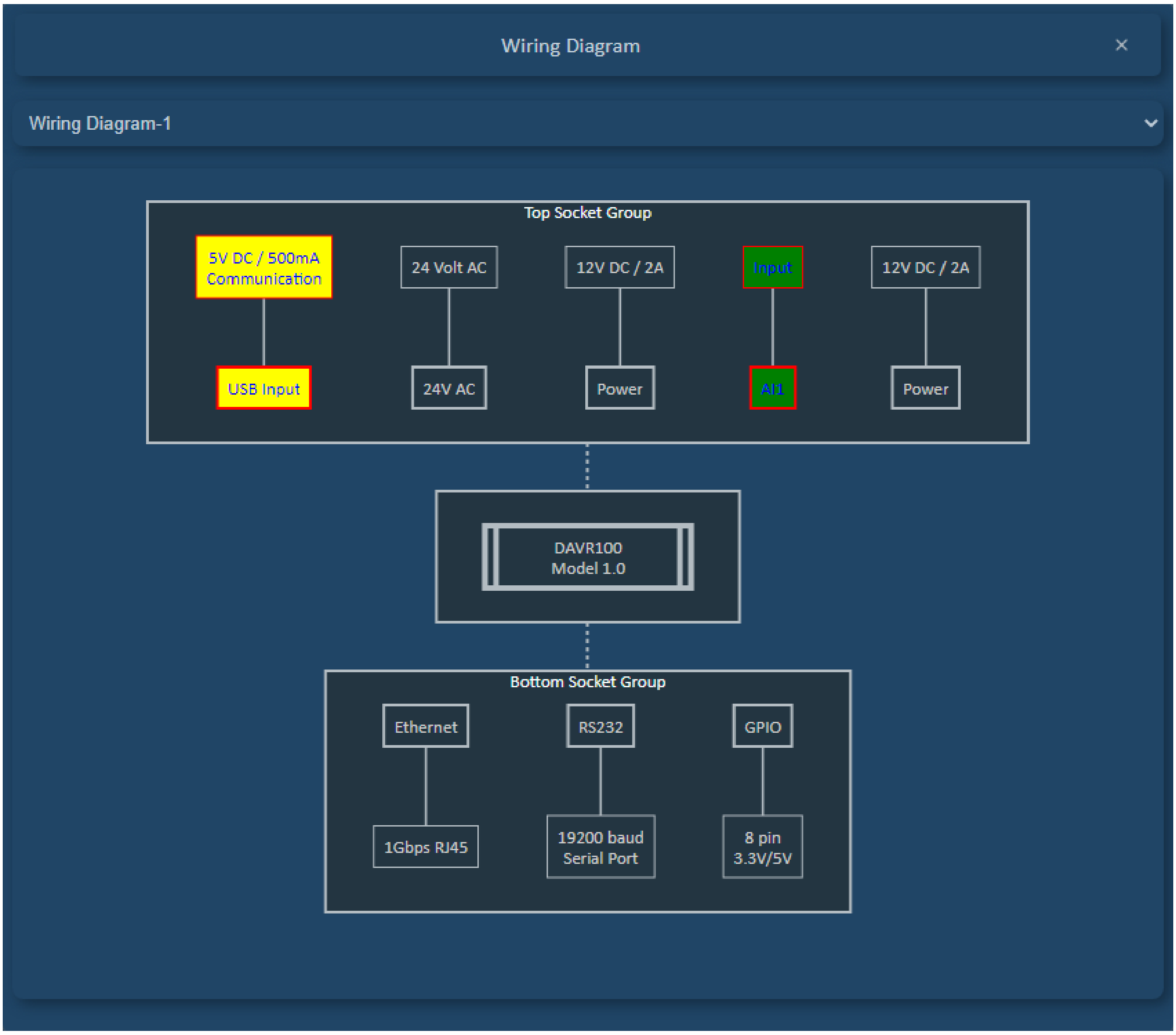
Important: Operation cannot be undone after pressing Apply All button. Parameters start being written to device. Therefore, carefully check changes on Configuration Review page.

9.6 Quick Configuration Advantages

- **Time Saving:** Set multiple parameters at once
- **Error Prevention:** Risk of incorrect configuration reduced with step-by-step guidance
- **Preview Capability:** View changes before applying
- **Bulk Application:** All parameters applied with single button
- **Reversible:** You can go back and make corrections at desired step
- **Profile Support:** Ready profiles for different scenarios

Tip: If setting up device for first time, you can quickly complete basic settings using Quick Configuration. Later you can make detailed adjustments from Parameter Settings page if necessary.

10. Wiring Diagram



The Wiring Diagram page is technical documentation page that visually shows your device's socket connections. With this page, you can make device connections correctly and prevent connection errors.

10.1 Wiring Diagram Overview

Page Features:

- Shows physical locations of device sockets
- Contains pin structure and numbering of each socket
- Explains functions and meanings of socket pins
- Visually represents connection schema

10.2 Connection Type Selection

If your device supports multiple connection configurations, you can select appropriate connection type from **Wiring Diagram** selection box.

Selection Box Features:

- Open **Wiring Diagram** selection box at top of page
- Available connection types displayed in list
- Select desired connection type
- **Examples:**
    - Standard Connection
    - Analog Input/Output
    - Special Configuration 1, 2, etc.

When you make selection, socket diagram specific to that connection type displays on screen.

10.3 Dynamic Diagram Feature

Wiring Diagram page can **change dynamically** according to your device's current parameter settings.

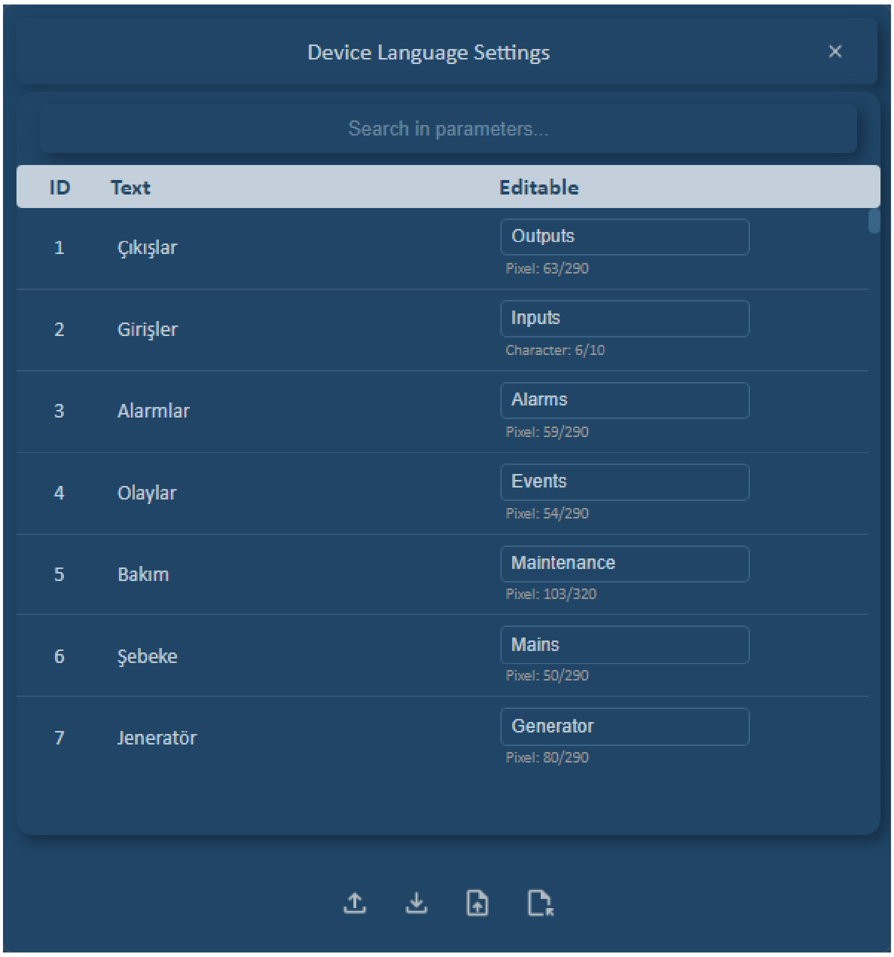
Dynamic Change Scenarios:

- If your parameter settings affect connection structure, diagram automatically updates
- **Example 1:** If Input/Output mode changed:
    - Functions of relevant pins updated
    - Connection schema suitable for new configuration displayed
- **Example 2:** If Active/Passive sensor selection made:
    - Power requirements and pin assignments change
    - Appropriate connection schema automatically loads

Important: After making parameter changes, check Wiring Diagram page to ensure your connections are still correct.

11. Device Language Settings Page

Device Language Settings page allows you to change language of texts displayed on device's own screen. With this feature, you can translate and customize texts on device screen to different languages.



Note: This page changes texts **on device screen**. To change Power Cloud MKII interface language, use Generate Language File page.

11.1 Opening Device Language Settings Page

Page Access:

Step 1: Click Device Language Button

- Press **"Device Language"** button from main page
- Device Language Settings page opens

Step 2: Loading Device Texts

- All **texts** in connected device's content automatically load
- Texts listed in table format

11.2 Device Language Settings Table

When page opens, table with two columns is displayed:

Column Descriptions:

Text Column:

- Shows **current text** on device
- Usually comes in default language (English)
- **Read-only** (cannot be edited)
- Used as reference
- Shows which text will be changed

Editable Column:

- Contains **editable** entry boxes
- Here you write expression you want corresponding text to have on device
- Each row can be edited separately
- Translation or customization written here
- Character/pixel limitations apply to this column

11.3 Editing Texts

Manual Editing:

Step 1: Find Text to Edit

- Find text you want to change in table
- Use search feature if necessary

Step 2: Click Editable Cell

- Click entry box in **Editable** column of relevant row
- Box enters edit mode

Step 3: Enter New Text

- Write corresponding text in desired language
- Pay attention to character/pixel limitations
- **Example:**
    - Text: "Start"
    - Editable: "Başlat" (Turkish)

Step 4: Move to Other Texts

- Click another box to proceed
- Continue editing next texts

11.4 Character and Pixel Limitations

Due to size constraints of device screens, maximum length limits exist for each text.

Limitation Types:

Two types of limitations may exist depending on device state:

- **Character Limit:** Maximum character count
- **Pixel Limit:** Maximum pixel width

Character Limit Example:

Indicator appears below entry box:

Character: 6/10

Meaning:

- Maximum **10 characters** can be written
- Currently **6 characters** written
- **4 more characters** can be written

Entry Behavior:

- When entry box reaches maximum character count, doesn't allow more typing
- Keyboard input blocked when limit reached

Pixel Limit Example:

Indicator appears below entry box:

Pixel: 50/200

Meaning:

- Maximum **200 pixel** width text can be written
- Current text using **50 pixels** width
- **150 pixels** more can be used

Pixel Calculation:

On some devices, text sizes measured in **pixels** instead of character count. In this case:

- **Font** is important
- **Font size** taken into account
- Each character can have different width (e.g., "i" narrow, "W" wide)
- Pixel value **calculated dynamically** during typing

11.5 Overflow and Approach Warnings

You receive visual warnings when approaching or exceeding limits.

When Overflow Occurs:

When limit exceeded:

- Entry automatically blocked
- You cannot type more characters
- You need to shorten text

Recommendation: When you see red warning, shorten text or use abbreviations. For example, write "Start" instead of "Starting".

11.6 Text Search

You can use search feature to quickly navigate through long text lists.

Using Search in Parameters:

Step 1: Type in Search Box

- Find **"Search in parameters..."** entry box at top of page
- Type word or word part you want to search

Step 2: View Results

- Table automatically filtered
- Only texts matching search criteria shown
- Search done in both Text and Editable columns

Step 3: Clear Filter

- When you clear search box, full list reappears

11.7 Automatic Translation Feature

Device Language Settings page offers automatic translation feature using online translate services.

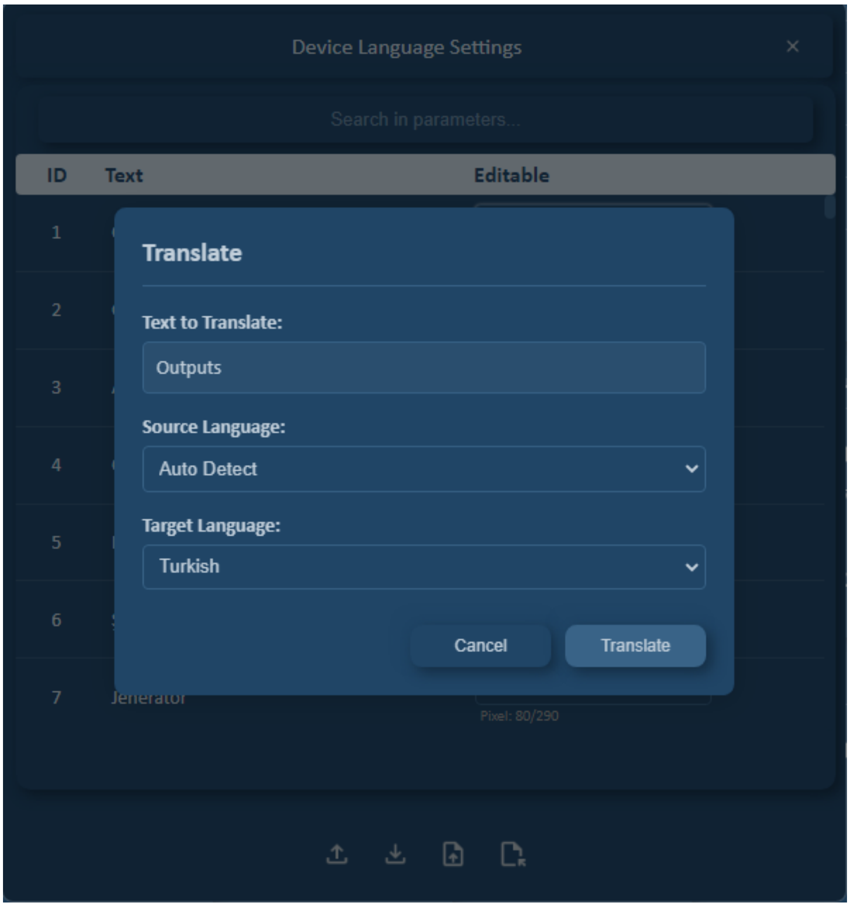
Using Automatic Translation:

Step 1: Open Translation Menu

- **Double-click** entry box in **Editable** column of text you want to translate
- Automatic translation menu opens

Step 2: Translation Menu Content

Following options in opened menu:



Menu Elements:

Source Language Selection Box:

- Select which language original text is in
- Device's default language (usually English) comes by default
- **Language options:**
  - Turkish
  - English
  - German
  - French
  - Chinese
  - Japanese
  - and other supported languages...

Target Language Selection Box:

- Select which language text will be translated to
- Select desired target language
- **Example:** English → Turkish

Translate Button:

- Starts translation process

Step 3: Select Languages

- **Source Language:** Select original text language (e.g., English)
- **Target Language:** Select target language (e.g., Turkish)

Step 4: Press Translate Button

- Click **Translate** button
- Online translation service called
- Text automatically translated
- Relevant entry box **filled with new translation**

Step 5: Check Translation

- Since automatic translation uses online translate services, may not always be perfect. Machine translation can make errors in context, technical terms, and idiomatic expressions.
- Check translated text
- Make manual corrections if necessary
- Check character/pixel limits

Note: Internet connection required for automatic translation. If no connection, you need to do manual translation.

11.8 Reading from Device and Writing to Device

Read From Device Button:

Used to read current texts in device.

Usage Scenarios:

- To see current translations if device previously translated to different language
- To check changes
- To refresh current state

Steps:

- Click **Read From Device** button
- Device queried
- Current texts read
- Editable column updated
- Translations saved on device shown

Warning: When you press Read From Device, unsaved changes you made on page will be lost. Make sure you save your changes first.

Set Language Button:

Used to write all edited texts to device.

Steps:

Step 1: Complete All Changes

- Review all texts in Editable column
- Check character/pixel limits
- Fix texts with overflow warnings

Step 2: Press Set Language Button

- Click **"Set Language"** button at bottom of page
- Writing process starts

Step 3: Monitor Writing Status

During process, writing status displayed on screen:

Status Indicators:

- Progress bar
- Completion percentage (%)

Step 4: Process Completion

- All texts written to device
- **"Successfully completed"** message displayed
- Device starts working with new texts

Step 5: Check Device

- Check device screen
- Verify texts appear correctly
- Make corrections if overflow or display problem

11.9 Exporting and Importing Language Files

Device Language Settings page offers feature to save and load language texts to file, similar to parameter export/import process.

Export File:

Step 1: Click Export File Button

- Press **"Export File"** button
- File save window opens

Step 2: Select File Format

Two format options:

Excel Format (.xlsx):

- Human-readable
- Suitable for documentation
- Two columns: Text and Translation
- All texts in table format

PRL Format (.prl):

- Power Cloud MKII proprietary format
- Used for Import operation
- For backup and restore

Step 3: Specify Save Location and Name

- Select folder where file will be saved
- Give file name (e.g., Device_Language_Turkish.prl)
- Press **Save** button

Export Usage Purposes:

- Backup: Backup current translations
- Sharing: Transfer translations to other devices
- Documentation: Keep translation records
- Version Control: Save different versions

Import File:

Used to load previously saved language file.

Step 1: Click Import File Button

- Press **"Import File"** button
- File selection window opens

Step 2: Select PRL File

- Find **.prl** extension language file you previously saved

- Select file and click **Open** button

Step 3: File Loads

- File automatically read
- **Editable** column filled with translations from file
- All texts updated

Step 4: Check and Edit

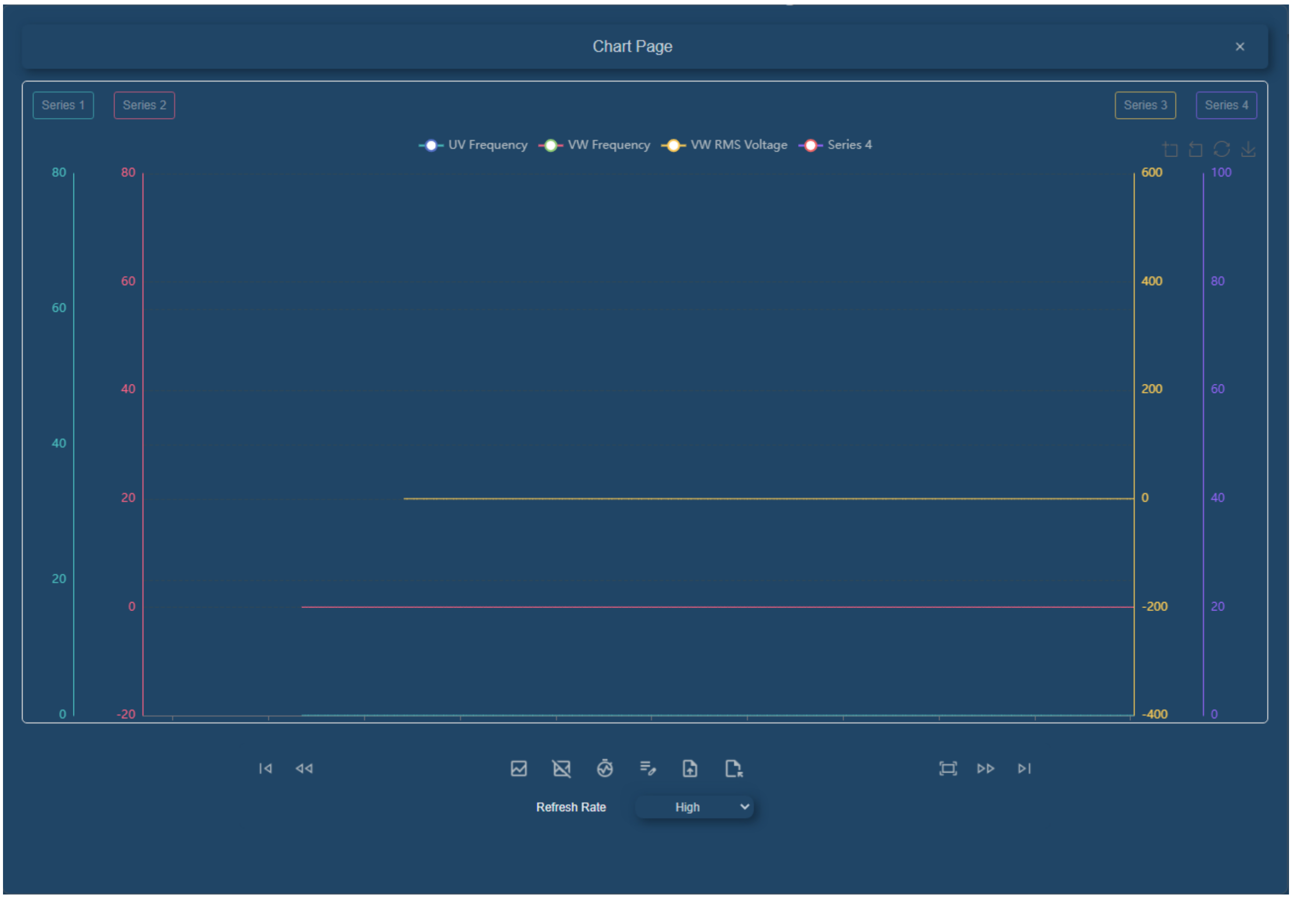
- Review loaded translations
- Make updates if necessary
- Check character/pixel limits

Step 5: Write to Device

- Write to device with Set Language button

12. Chart Page

The Chart (Graph) page allows you to visualize data from device in real-time. In line chart format, you can monitor, compare, and analyze up to 4 different parameters simultaneously.



12.1 Opening Chart Page

Page Access:

Step 1: Click Chart Button

- Press **"Chart"** button from main page
- Chart page opens

Step 2: Chart Area Displayed

- Empty chart screen shown
- Series selection buttons become active
- Control buttons ready to use

12.2 Chart Page General Structure

Chart page consists of these main sections:

Vertical Scale Lines:

- 2 scale lines **on left** of page (for Series 1 and Series 2)
- 2 scale lines **on right** of page (for Series 3 and Series 4)

- Value indicators on each scale line
- Scale lines show parameter's min/max values

Series Buttons:

- Series button above each scale line
- From left to right: **Series 1, Series 2, Series 3, Series 4**
- These buttons used to select parameters

Chart Area:

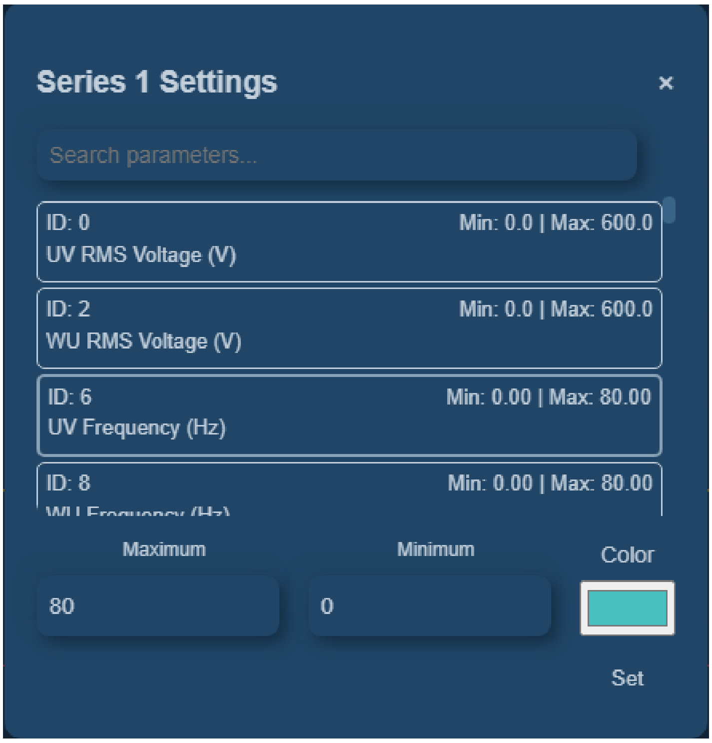
- Large chart drawing area in center
- Time axis (horizontal) and value axes (vertical)
- Line charts of selected parameters
- Real-time data flow

Control Buttons:

- Control buttons below and above chart
- Play, stop, zoom, export, etc.

12.3 Series (Parameter) Selection

Each series button allows you to select parameter to be shown on chart.



Series Selection Steps:

Step 1: Click Series Button

- Click series button you want to configure (e.g., **Series 1**)
- **Series Settings** page opens

Step 2: Series Settings Page Content

Following sections in opened page:

Parameter List:

- List of parameters in device
- Parameter names and ID numbers
- Scrollable list format

Search Parameters Entry Box:

- Located at top of list
- Used to search parameters
- **Search by name:** "Motor", "Temperature", "Voltage"
- **Search by ID:** "100", "250", "305"
- List automatically filtered as you type

Step 3: Parameter Selection

- Select desired parameter from list
- Example: "Motor Speed", "Temperature", "Voltage"

Step 4: Scale Settings (Minimum and Maximum)

Two entry boxes below list:

Minimum Value:

- Lower limit of chart scale
- **Default:** Parameter's defined minimum value
- User can change this value
- **Example:** 0, -50, 100

Maximum Value:

- Upper limit of chart scale
- **Default:** Parameter's defined maximum value
- User can change this value
- **Example:** 100, 200, 500

Note: Min/Max values determine scale range on chart's vertical axis. Parameter's actual values visualized in this range.

Step 5: Color Selection

Color (Color Palette):

- Color selection box
- Determines which color this series' line chart will show in
- Select desired color from color palette

Step 6: Save Settings

- Press **"Set"** button after making all settings
- Series settings saved
- Page closes
- Selected parameter starts appearing on chart

Step 7: Set Other Series

- Repeat same process for **Series 2, Series 3, Series 4**
- You can select different parameter for each series
- You can assign different color and scale values

12.4 Parameter Indicators and Show/Hide

At top of chart, selected parameters shown with colored dots.

Parameter Indicators:

4 dots and parameter names appear above chart:

Each indicator contains:

- **Colored dot:** Series color
- **Parameter name:** Name of selected parameter

Show/Hide Feature:

Hiding Parameter:

- **Click** on relevant parameter indicator (colored dot)
- That parameter's line **disappears** on chart
- Scale line remains visible
- Parameter indicator appears dim/gray

Showing Parameter Again:

- **Click again** on same parameter indicator
- Line **reappears** on chart
- Parameter indicator returns to normal color

Usage Scenarios:

Scenario 1: Detailed Analysis

- 4 parameters selected
- Hide Motor Speed and Temperature
- Examine only Voltage and Current
- Detailed correlation analysis

Scenario 2: Reduce Complexity

- Chart looks too complicated
- Hide parameters not currently relevant
- Focus only on data of interest

Scenario 3: Screenshot

- Taking chart for report
- Hide unnecessary parameters
- Get clean chart showing only important data

> **Advantage:** With Show/Hide feature, you can do quick analysis by changing only visibility without changing chart settings.

12.5 Chart Control Buttons

Main control buttons at bottom of chart:

Start Chart

- **Starts** real-time data flow
- Chart starts flowing from right to left
- Data updates live
- **Usage:** For continuous monitoring

Stop Chart

- **Stops** real-time data flow
- Chart freezes
- You can analyze on current view
- **Usage:** For detailed examination

Time Line

- Makes timeline bar **visible**
- Timeline bar appears below chart
- **When pressed again:** Timeline hides
- **Usage:** For navigating through time

Export File

- Exports current **data on chart** in Excel format
- All series data saved in table format
- **Format:** .xlsx (Excel)
- **Content:** Timestamp and all parameter values

Export Configuration

- Exports selected **series settings**
- **Saved information:**
    - Parameter selections
    - Min/Max values
    - Color settings
- **Usage:** Backup and share settings

Import Configuration

- Loads previously exported **configuration file**
- All series settings automatically applied
- No need for manual setup each time
- **Usage:** Quick setup and standard views

12.6 Chart Navigation Buttons

Go to Start

- Brings chart to **initial starting point**
- Shows moment when data flow started
- To examine all recorded data from beginning

Move Backward

- Moves chart **one view page earlier**
- You can go back step by step
- To examine past data

Move Forward

- Moves chart **one view page forward**
- You can advance step by step
- To see next data in stopped chart

Go to End

- Brings chart to **last point**
- Shows most recent data
- To return to real-time state

Fit All Chart

- Automatically **scales to fit all data on one screen**
- Overall view of all data from start to end
- Ideal for trend analysis

12.7 Chart Flow Direction and Interaction

Chart Flow Direction:

Chart flows **from right to left**:

- New data enters from right side
- Old data slides to left
- Data exiting from left disappears from view (but remains in memory)
- Real-time data flow continues continuously

Flow Control:

- **Start Chart:** Start/continue flow
- **Stop Chart:** Stop flow
- You can navigate on stopped chart

12.8 Cursor Modes

Two different cursor modes on chart:

PAN MODE (Drag and Scroll):

- **Default mode**
- Cursor is normal arrow shape
- **Mouse usage:**
  - **Click and Drag:** Slides chart left-right
  - Start and end points change
  - You can navigate forward-backward through time
- **Mouse Scroll usage:**
  - **Scroll up:** Slides chart left (goes back)
  - **Scroll down:** Slides chart right (goes forward)

ZOOM MODE:

- Activated by pressing **"Zoom"** button in upper right corner

Zoom with Mouse:

- **Click** on chart
- **Hold and drag** to right
- Selected area highlighted as rectangle
- When **released**, that area zoomed (horizontal zoom)
- Only zooms on **horizontal (time)** axis

Zoom with Mouse Scroll:

- **Scroll up:** Zoom in
- **Scroll down:** Zoom out
- Zooms centered on point where cursor is located

Exiting Zoom Mode:

- Press **Zoom** button again
- Cursor automatically returns to **Pan mode**

Zoom Reset:

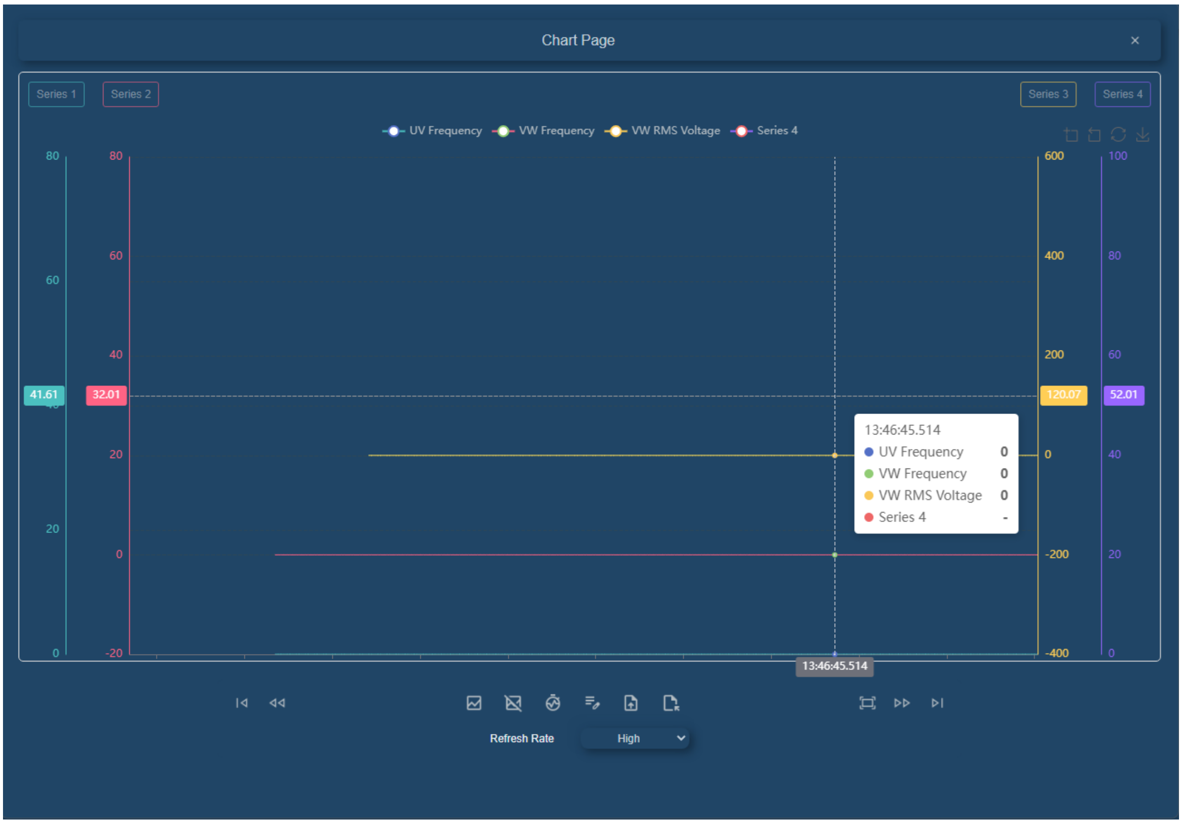
- Press **"Zoom Reset"** button in upper right corner
- All zoom operations canceled
- Chart returns to original scale
- Default view restored

Zoom and Pan Comparison:

Feature	Pan Mode	Zoom Mode
Default	✓ Yes	✗ No
Scroll	✓ Left/Right	✗ Cannot
Zoom	✗ Cannot	✓ Yes
Mouse Scroll	Scrolls	Zooms
Drag-Drop	Scrolls	Selects zoom area

Value Display (Tooltip):

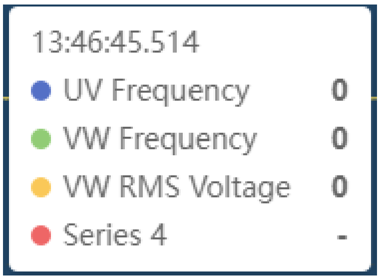
In both modes (Pan and Zoom), when you navigate on chart with cursor, **value display box** automatically opens.



Features:

- Shows **all parameter values** at point where cursor is located
- **Timestamp** (date and time) displayed
- Each parameter shown with its own **color**
- Values **dynamically update** as cursor moves
- Box disappears when you exit chart

Value Box Example:



Usage:

- To see values at specific point on chart, bring cursor to that point
- Value box automatically appears
- Navigate on chart to examine values at different time points
- Ideal for precise data reading and analysis

Tip: Using value box, you can manually read values at any point on chart. This feature very useful for trend analysis and anomaly detection.

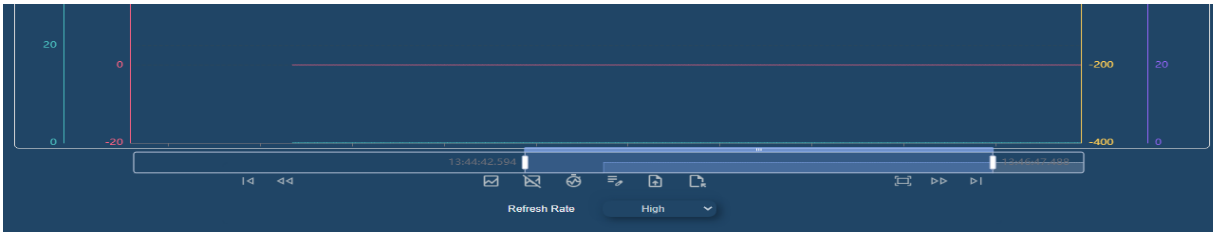
12.9 Additional Chart Features

Save as Image:

- Press **"Save as Image"** button in upper right corner
- **Screenshot** of chart taken
- Saved as image file in **PNG** format
- Ideal for reporting and documentation

Timeline Bar:

- Activated with **Time Line** button
- Appears **below** chart
- Shows all data range as small preview
- **Mouse usage:**
  - **Move forward-backward** through time by dragging bar
  - Displayed area highlighted
  - Corresponding time period shown on main chart



Refresh Rate:

Refresh Rate selection box to adjust chart refresh speed:

Options:

- **Low:** Slow update
- **Medium:** Standard update
- **High:** Fast update

Note: High refresh rate provides smoother chart but uses more system resources. Select optimal value according to your needs.

12.10 Chart Usage Scenarios

Scenario 1: Real-Time Monitoring

1. Select 4 parameters you want to monitor
2. Refresh Rate: High
3. Press Start Chart button
4. Monitor live changes of parameters
5. Stop Chart if you see abnormality

Scenario 2: Historical Data Analysis

1. Stop with Stop Chart
2. Go to start with Go to Start
3. Activate zoom mode
4. Select and zoom time period of interest
5. Examine in detail with pan mode
6. Take screenshots of important parts

Scenario 3: Comparative Analysis

1. Select 4 related parameters (e.g., Voltage, Current, Power, Temp)
2. Set in different colors
3. See all data with Fit All
4. Examine correlations
5. Do detail analysis by hiding irrelevant parameters

Scenario 4: Report Preparation

1. Save settings with Export Configuration
2. Collect data with Start Chart
3. Capture critical moments with Stop Chart
4. Take screenshots with Save as Image
5. Export data to Excel with Export File
6. Use in report

Scenario 5: Share Standard View

1. Make optimal chart settings
2. Save with Export Configuration
3. Share file with team members
4. Everyone uses same view with Import Configuration
5. Consistent monitoring and reporting

12.11 Troubleshooting

Problem 1: "Chart Not Updating"

- **Solution:**
  - Press Start Chart button
  - Check device connection
  - Check Refresh Rate setting
  - Ensure parameters not hidden

Problem 2: "Lines Not Visible"

- **Solution:**
  - Make visible by clicking parameter indicators
  - Check scale min/max values
  - Verify data is in scale range
  - Redo series settings

Problem 3: "Zoom Not Working"

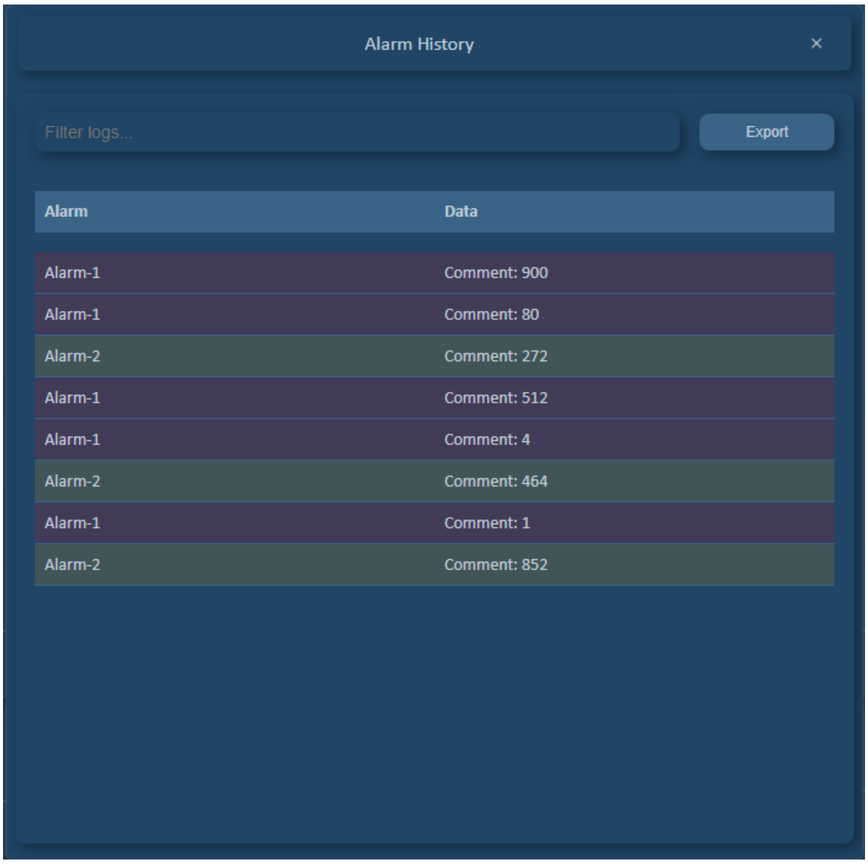
- **Solution:**
  - Ensure Zoom button pressed
  - Click and drag to right (not left)
  - Reset with Zoom Reset and try again

Problem 4: "Cannot Import Configuration"

- **Solution:**
  - Check file is in correct format
  - Ensure file path is accessible
  - Try different configuration file
  - Repeat export process

13. Alarm/Event History

The Alarm History page allows you to view records of alarms that occurred on device. With this page, you can examine, analyze, and report past alarm events.



13.1 Opening Alarm History Page

Page Access:

Step 1: Click Alarm History Button

- Press **"Alarm History"** button from main page
- Alarm History page opens

Step 2: Loading Alarm Records

- **Alarm history information** in device automatically loads
- All alarm records listed in table format

13.2 Alarm History Table

When page opens, table with two columns is displayed:

Column Descriptions:

Alarm Column:

- Shows **alarm name**
- Indicates which alarm event occurred
- Short and descriptive alarm names
- **Examples:**
  - "Motor Overload"
  - "High Temperature"

- "Communication Error"
- "Low Voltage"
- "Emergency Stop"
- "Sensor Fault"

Data Column:

- Shows explanatory **information** related to alarm

13.3 Alarm Severity Color Coding

In Alarm History table, rows automatically color-coded according to **alarm severity**. This way you can quickly notice critical alarms.

13.4 Alarm Search and Filtering

You can use search feature to quickly find specific alarm in long alarm lists.

Using Filter:

Step 1: Type in Filter Entry Box

- Find **"Filter log..."** entry box at top of page
- Type word or word part you want to search

Step 2: View Results

- Table automatically filtered
- Only alarms matching search criteria shown
- Search done in both Alarm and Data columns

Step 3: Clear Filter

- When you clear search box, full list reappears

13.5 Exporting Alarm History

You can export alarm records for documentation, analysis, or reporting purposes.

Export Process:

Step 1: Click Export Button

- Press **"Export"** button on page
- File save window opens

Step 2: Specify Save Location and Name

- Select **folder** where file will be saved
- Specify **file name**
- Suggested format: Alarm_History_2025_10_06.csv

Step 3: Save in CSV Format

- File automatically saved in **CSV** format
- Press **Save** button
- Export process completes

CSV File Features:

Using CSV File:

- Can be opened with **Microsoft Excel**
- Can be viewed with **Google Sheets**
- Can be processed in **data analysis** programs
- Can be added to **reports**

- Can be used for **chart creation**
- Can be saved for **archiving**

Export Usage Purposes:

1. Documentation:

- For maintenance reports
- For technical documentation
- For customer reports

2. Analysis:

- Analyze alarm trends
- Identify most frequent alarms
- Time-based analysis
- Statistical evaluation

3. Archiving:

- Long-term record keeping
- Warranty coverage documentation

4. Sharing:

- Share with technical team
- Communication with manufacturer
- Inform maintenance team

NOTE: Event History page also works same structure as Alarm History. Event History records system events that are not alarm-natured. Event History page also offers two-column table (Event and Data), filtering feature, and CSV export capability. Only difference is that events are generally informational.

14. Background Image Page

Background Image page is used on devices that support customizing startup screen image. On this page, you can customize device's startup screen, background color, progress bar color, and QR code content.

Note: This feature only available on device models that support changing startup screen image. If your device doesn't support this feature, this page won't appear in menu.

14.1 Background Image Page Overview

Four main customization options on page:

- **Background Image:** Image to be shown at device startup
- **Background Color:** Background color of startup screen
- **Progress Color:** Color of loading bar during device startup
- **QR Code Content:** Text QR code on device screen will contain

14.2 Selecting Startup Image

Image Selection Methods:

Two different ways to select image on Background Image page:

Method 1: Double Click

- **Double-click** on area that says **"Please select an image"** on page
- File selection window automatically opens

Method 2: Select Image Button

- Click **"Select Image"** button on page
- File selection window opens

Image File Selection:

- Find your image file in opened file selection window
- Supported formats:
  - **JPG/JPEG** (.jpg, .jpeg)
  - **PNG** (.png)

    - **BMP** (.bmp)
- Select appropriate image and click **Open** button

Displaying Selected Image:

- After image selected, automatically appears in **image panel**
- Selected image shown as preview according to device screen dimensions
- You can check in this panel how image will appear on device screen
- You can change by selecting different image if desired

Image Selection Recommendations:

- Use image with size suitable for device screen resolution
- Prefer high-contrast and clear images
- Simple and understandable designs look better
- Images containing logo or company information can be used

14.3 Setting Background Color

Background Color option determines background color of device startup screen.

Color Selection:

- Click color box labeled **"Background Color"** on page
- Select desired color with opened **color palette**
- In color palette:
    - You can select from ready colors
    - You can enter custom color with RGB or HEX values
    - You can adjust color tones

Background Color Usage Areas:

- Visible in areas where image doesn't exist
- Shown as background color during loading
- Brand-appropriate color selection can be made

Color Selection Recommendations:

- Select color compatible with image
- Ensure progress bar color visible on background
- Corporate colors can be used

14.4 Setting Progress Bar Color

Progress Color option determines color of loading/progress bar during device startup.

Color Selection:

- Click color box labeled **"Progress Color"** on page
- Select desired color with opened **color palette**

Progress Bar Features:

- Displayed during device startup process
- Visually shows loading progress
- Should be contrasted with background color
- Shows user device is loading

Progress Color Selection Recommendations:

- Provide high contrast with background color
- Should be clearly visible on background

Example Combinations:

- Dark background + Light progress bar (e.g., Black + White)
- Light background + Bright progress bar (e.g., White + Green)
- Blue background + Yellow progress bar (high contrast)

14.5 Setting QR Code Content

QR Code Content feature allows you to determine information QR code shown on device screen will contain.

QR Code Content Entry:

- Find **"QR Code Content"** entry box on page
- Write text you want in box
- Entered text creates QR code information in device

Content Types Can Be Entered:

- **Web Address (URL):** https://www.company.com/device-detail
- **Email Address:** info@company.com
- **Phone Number:** +90 5xx xxx xx xx
- **Plain Text:** Device Serial No: ABC123456
- **vCard Information:** Contact card information

QR Code Usage:

- QR code shown on device screen
- When scanned with **QR code reader**, entered text/information displayed
- Can be read with mobile device cameras or QR code applications
- Used for quick information access and documentation

QR Code Usage Scenarios:

- **Device Information:** Serial number, model information
- **Technical Support:** Support webpage or phone number
- **User Manual:** Online manual link
- **Warranty Information:** Warranty inquiry page
- **Company Contact:** Company website or contact information
- **Video Tutorials:** Video or training platform link

QR Code Content Recommendations:

- Enter short and meaningful content (long texts make QR code complex)
- If using URL, ensure starts with https://
- Test content (check with QR code reader)
- Enter current information (unchangeable information should be preferred)

Note: Longer QR code content, more complex and dense QR code becomes. This can make readability difficult. Use short URLs or reference codes when possible.

14.6 Loading Settings to Device

After making all customizations (image, colors, QR code), you need to load them to device.

Loading Steps:

Step 1: Check Settings

- Check selected image
- Review Background Color and Progress Color
- Verify QR Code Content

Step 2: Press Load Button

- Click **"Load"** button at bottom of page
- Loading process starts

Step 3: Monitor Loading Process

During loading, following information displayed on screen:

- **Progress Indicator:** Loading status shown as percentage (%)

Step 4: Loading Completion

- When loading reaches 100%, process completes
- **"Loading successful"** message displayed
- **Device connection automatically disconnects**
- This is normal and expected behavior

Step 5: Reconnect

- You need to **reconnect** since connection disconnected
- Press **Connect** button on main page to reconnect
- New startup screen and settings now active

Important: Don't cut power or disconnect device during loading process. Wait until process completes.

14.7 Post-Loading Check

Verifying Changes:

- Restart device
- Check new image and colors on startup screen
- Verify progress bar appears in correct color
- Test QR code by scanning with reader

QR Code Test:

- Open QR code reader on mobile device
- Scan QR code on device screen
- Check entered content appears correctly
- If web address, test redirects to correct page

14.8 Background Image Usage Scenarios

Scenario 1: Corporate Brand Identity

- Use company logo as startup image
- Set corporate colors as background and progress color
- Add company website to QR code

Scenario 2: Device Information Screen

- Custom graphic containing device model

- Link to technical specifications page with QR code
- Compatible colors for professional appearance

Scenario 3: Technical Support Access

- Support contact information on startup screen
- Direct redirect to support form with QR code
- Attention-grabbing design with bright progress bar

Scenario 4: User Manual Access

- Simple and understandable startup graphics
- Link to online user manual with QR code
- User-friendly color palette

Tip: Testing Background Image customizations on test device before applying to serial production or all devices helps prevent possible problems.

15. Firmware Update

Firmware Update page is used to update your device's software. Power Cloud MKII offers two different loading methods, allowing firmware upload to both devices with existing software and completely empty devices.

15.1 Firmware Update Methods

Two different firmware loading methods in Power Cloud MKII:

- **Method 1:** If device has existing software (Normal Update)
- **Method 2:** If device has no software (Initial Loading / Bootloader Mode)

Note: Which method to use depends on your device's current state. If device working and can establish connection, use Method 1, otherwise use Method 2.

Method 1: Update with Existing Software

This method used when device has existing software and connection can be established with Power Cloud MKII.

Step-by-Step Loading Process:

Step 1: Connect to Device

- Start Power Cloud MKII
- Connect to device by pressing Connect button
- Ensure device successfully recognized

Step 2: Press Firmware Update Button

- Click **Firmware Update** button from main page
- File selection window automatically opens

Step 3: Select Firmware File

- Find firmware file with **.blf** extension in opened file selection screen
- Select file and press **Open** button
- Verify file name and location

Important: Only files with **.blf** extension are valid firmware files. Be careful not to select wrong or damaged file.

Step 4: Automatic Reset and Loading Start

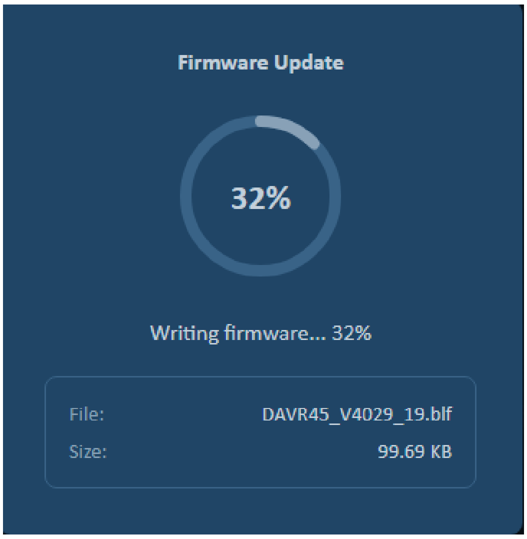
- After firmware file selected, device **automatically resets**
- Device enters bootloader mode
- Firmware loading process automatically starts

Step 5: Monitor Loading Process

During loading, following information displayed on screen:

- **Progress Indicator:** Loading progress shown as **percentage (%)**
- **File Name:** Name of firmware file being loaded
- **File Size:** Total size of firmware file
- **Status Message:** Status information like "Writing...", "Successfully!" etc.

Example View:



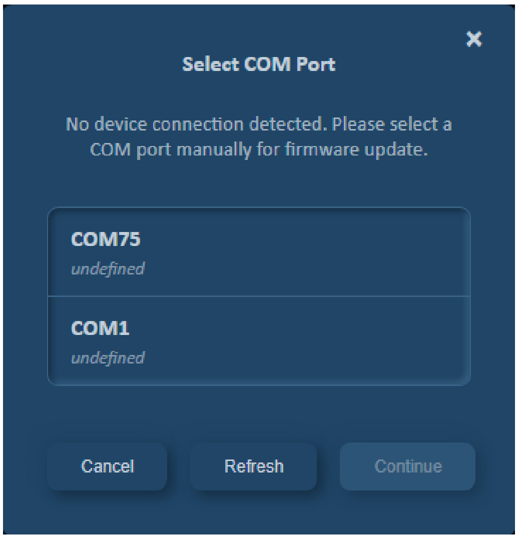
Step 6: Loading Completion

- When loading reaches 100%, process completes
- Success message displayed
- Page automatically closes
- Device restarts with new software

Warning: Never cut power or disconnect during firmware loading process.

Method 2: Initial Loading to Empty Device (Bootloader Mode)

This method used when device has no software or existing software corrupted. In this case, normal connection to device cannot be established.



Step-by-Step Loading Process:

Step 1: Press Firmware Update Button (Without Connection)

- In Power Cloud MKII, click Firmware Update button **without connecting to device yet**
- Port selection window opens

Step 2: View Port List

- Opened window displays **port list** connected to computer
- List shows COM port numbers (COM1, COM3, COM5, etc.)
- Determine port device is connected to

Refreshing Port List:

- If port you're connected to doesn't appear in list:
  - Press **Refresh** button
  - Port list rescanned and updated
  - Newly connected ports become visible in list

Tip: If unsure which port is correct, unplug device and plug back in, then press Refresh button. Newly appearing port is port your device is connected to.

Step 3: Port Selection

- Select port your device is connected to from list
- Be sure of your selection

Step 4: Press Continue Button

- Click **Continue** button after making port selection
- File selection window opens

Step 5: Select Firmware File

- Find firmware file with **.blf** extension in opened window
- Select file and click **Open** button

Step 6: Firmware Loading Starts

- Loading automatically starts after file selected
- No reset required since device in bootloader mode
- Loading process starts directly

Step 7: Monitor Loading Process

Following information displayed on screen as in Method 1:

- **Progress Indicator:** Progress status shown as percentage (%)
- **File Name:** Name of firmware file being loaded
- **File Size:** File size information
- **Status Messages:** Real-time status updates

Step 8: Loading Completion

- When loading reaches 100%, process completes
- Success message shown
- Page automatically closes
- Device starts working with new software
- Can now establish normal connection

15.2 Precautions During Firmware Loading

Security Measures:

- **Don't Cut Power:** Never cut device power or USB connection during loading

- **Select Correct File:** Ensure using appropriate firmware file for your device
- **Don't Disconnect:** Don't remove USB cable or serial port connection

15.3 Firmware Loading Problems and Solutions

Problem 1: "Port Not Found"

- **Solution:**
    - Ensure device properly connected to computer
    - Unplug and replug USB cable
    - Press Refresh button to refresh port list
    - Try different USB port if necessary

Problem 2: "Loading Not Starting"

- **Solution:**
    - Ensure selected correct firmware file (.blf extension)
    - Check file not damaged
    - Ensure device in bootloader mode

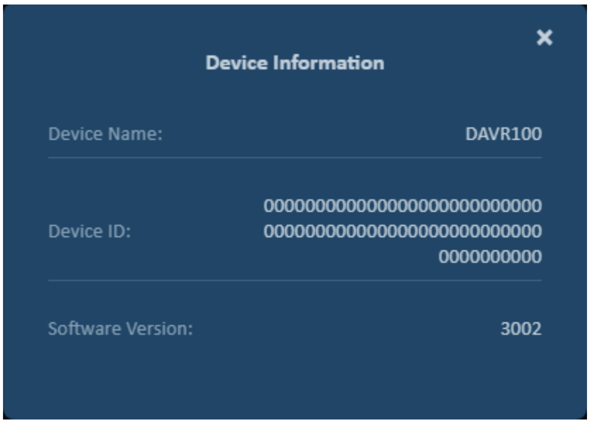
Problem 3: "Loading Stopped Midway"

- **Solution:**
    - Check connection
    - Restart process from beginning
    - Try different USB cable
    - Use Method 2 if necessary

Technical Support: If encounter unsolvable problem during firmware loading, note device model information, firmware file name, and error message and contact technical support team.

16. Device Information Page

Device Information page allows you to view information about connected device. This page contains important information for device identification, version control, and support processes. .



16.1 Opening Device Information Page

Access from Main Page:

To access Device Information page, click device information in lower right corner of main screen.

Clickable Device Information:

In lower right corner of main page, following information displayed and **clickable**:

Displayed Information:

On main page, following information appears abbreviated:

- **Device Name:** Model or name of connected device (e.g., Device X)
- **Version:** Device software version (e.g., v1000)
- **User Level:** User information (e.g., Operator/Factory)

16.2 Device Information Page Content

Device Information page displays following information about connected device:

Device Information:

1. Device Name

- Device model name

2. Device ID

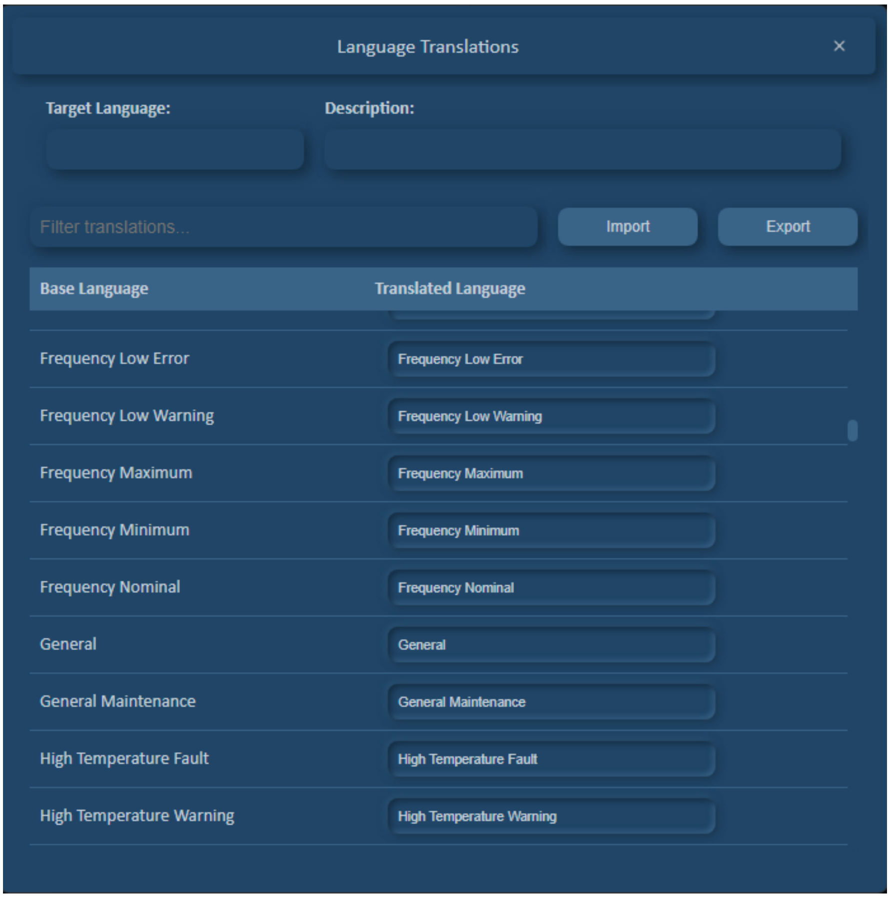
- **Unique** identification number specific to each device
- Unique identifier distinguishing device from other devices
- Fixed value assigned at factory
- Cannot be changed and unique for each device

3. Software Version

- Firmware/software version installed on device

17. Generate Language File

Generate Language File page allows you to create language file to display parametric and device-specific texts used in Power Cloud MKII software in different languages. With this feature, you can translate software to your own language or edit existing translation.



17.1 Opening Language Translations Page

Step 1: Click Generate Language File Button

- Press **"Generate Language File"** button from main page
- **Language Translations** page opens

Step 2: Loading Texts

- **All texts** in connected device automatically load
- Texts listed in table format
- Both parametric texts and device-specific texts included

Important: To use this feature, you must be connected to device. If no device connection, texts cannot be loaded. Only previously created file can be opened.

17.2 Language Translations Table

When page opens, table with two columns is displayed:

Column Descriptions:

Base Language:

- Contains **default** texts in device
- Comes in **English**
- This column **cannot be edited** (read-only)
- Used as reference
- Shows which text will be translated

Translated Language:

- **Editable** column
- By default comes with same values as Base Language
- User can update this column in desired language
- Each row can be edited separately
- Translation written here

Default State:

- When page first opened, both columns have **default values** (English)
- User translates Translated Language column to own language

17.3 Translating Texts

Translation Process Steps:

Step 1: Find Text to Translate

- Find text you want to translate in table
- Can use **Filter translations...** entry box in long lists

Step 2: Click Translated Language Cell

- Click cell in **Translated Language** column of relevant row
- Cell enters edit mode

Step 3: Enter Translation

- Delete existing text or write over it
- Enter corresponding text in desired language
- **Example:**
    - Base Language: "Motor Speed"
    - Translated Language: "Motor Hızı" (Turkish)

Step 4: Move to Other Texts

- Continue translation of next texts

Translation Recommendations:

- **Consistency:** Translate same terms same way
- **Brevity:** Keep close to original length when possible
- **Control:** Pay attention to spelling errors
- **Context:** Translate thinking where text will be used

17.4 Text Search and Filtering

You can use search feature to quickly navigate through long text lists.

Using Filter Translations:

Step 1: Type in Search Box

- Find **"Filter translations..."** entry box at top of page
- Type word or word part you want to search

Step 2: View Results

- Table automatically filtered
- Only texts matching search criteria shown
- Search done in both Base Language and Translated Language columns

Step 3: Clear Filter

- When you clear search box, full list reappears

17.5 Exporting Language File

After completing translations, you need to save them as language file.

Export Process Steps:

Step 1: Enter Target Language Information

- Type target language information in **"Target Language"** entry box on page
- **Examples:**
    - "Turkish" (English format)
    - "Türkçe"
    - "tr-TR" (ISO code)
    - "Español"
    - "Deutsch"

Step 2: Enter Description Information (Optional)

- Can add descriptive information in **"Description"** entry box
- **Examples:**
    - "Turkish language pack - v1.0"
    - "Power Cloud MKII Turkish translation"
    - "Turkish language file for ENKO Devices"
    - "Creation date: 03.10.2025"

Note: Description field optional but useful for file management. Especially if creating multiple versions, adding description recommended.

Step 3: Press Export Button

- Click **"Export"** button
- File save window opens

Step 4: Select Save Location

- Select **folder** where file will be saved in opened window
- Specify **file name**
- Suggested format: PowerCloud_Turkish.json or language_tr.json
- File format automatically set as **JSON**

Step 5: Click Save Button

- Press **Save** button
- Language file saved to location you selected

Important: This file becomes language file for Power Cloud MKII. You can use it as new language option in software by copying file to appropriate folder.

17.6 Importing Language File

You can open previously saved language file to view or edit translations.

Import Process Steps:

Step 1: Click Import Button

- Press **"Import"** button on Language Translations page
- File selection window opens

Step 2: Select JSON File

- Find **.json** extension language file you previously saved
- Select file and click **Open** button

Step 3: File Loads

- JSON file automatically read
- **Translated Language** column filled with translations from file
- **Base Language** column remains with original texts from device
- **Target Language** and **Description** fields loaded from file

Step 4: Make Edits

- Review loaded translations
- Make updates if necessary
- Fix incorrect translations

Step 5: Export Again

- After completing your edits
- Save updated file with Export button
- Can increase version number (e.g., v1.0 → v1.1)

Import Usage Scenarios:

- **Continue Translation:** Complete unfinished translation
- **Update:** Update existing translation with new texts
- **Correction:** Fix errors or wrong translations
- **Version Control:** Compare different versions
- **Backup:** Open and check old version
- **Copy:** Create similar language based on one language

17.7 Best Practices and Tips

For Translation Process:

- **Categorize:** Find and translate similar texts in groups using filter
- **Consistency:** Create terminology list and make consistent translations
- **Regular Save:** Export at certain intervals to save your progress
- **Backup:** Save with different names at each important stage

For File Management:

- **Version Control:** Add version to file name (e.g., Turkish_v1.2.json)
- **Add Date:** Add date to file name (e.g., Turkish_2025_10_03.json)
- **Use Description:** Always fill Description field

For Quality Control:

- **Test:** Test language file you created in software
- **Screen Check:** Check how translations appear on screen

- **Length Check:** Check if too long translations get cut off
- **Context Check:** Verify texts meaningful where they're used

Multi-Language Management:

- If creating multiple languages, create separate folder for each language
- Keep common terminology consistent across all languages
- Main language (usually English) always reference

IMPORTANT NOTE: If re-importing previously created language file, you must connect with default language (English) selected. Otherwise, your previous translations may be deleted or overwritten. This is because Base Language field comes from device's default values. If you connect in different language, Base Language column will be filled with that language's translations and original reference will be lost.

18. Using Language File in Power Cloud MKII

You can view interface and device texts in your own language by using language file you previously created in Power Cloud MKII software. Language selection made before connecting to device and your selection saved.

18.1 Language Selection and Connecting to Device

Language Selection Box:

When you start Power Cloud MKII, **language selection box** appears on main screen before connecting to device.

Language Options:

Two options in language selection box:

- **Default:** Software appears in default language (English)
- **User Define:** Custom language file created by user used

18.2 Connecting with Default Language

Using Default Option:

Step 1: Select Default Option

- Select **"Default"** option from language selection box
- Software will appear in English

Step 2: Press Connect Button

- Click **Connect** button to connect to device
- All interface and device texts shown in English

Note: Default option uses software's original English interface. This ensures all texts displayed in standard form.

18.3 Connecting with User Define (Custom Language File)

Using Custom Language File:

Step 1: Select User Define Option

- Select **"User Define"** option from language selection box
- File selection window automatically opens

Step 2: Select Language File

- Find language file you previously created in opened file selection window
- Select file and click **Open** button

Step 3: Press Connect Button

- After language file selected, click **Connect** button
- Connection established to device
- Now Power Cloud MKII appears with translations in language file you selected
- All interface texts and device parameters shown in language you determined

Step 4: Check Translations Active

- Check interface texts
- Review parameter names
- Ensure button and menu texts appear correctly

Success: Power Cloud MKII now working with completely customized texts with language file you prepared!

18.4 Language Selection Box After Connection

Important Behavior:

- After successfully connecting to device, **language selection box disappears**
- Language cannot be changed while connection active
- Language selection area becomes invisible

If Want to Select Different Language:

To change language after connection established:

- Press **Disconnect** button to close current connection
- Language selection box becomes visible again
- Select different language option (Default or another User Define file)
- Press **Connect** button again to connect with new language
- Software appears with selected new language

18.5 Saving Language Selection

Power Cloud MKII **automatically saves** your last language selection.

Auto-Save Features:

- Language you selected (Default or User Define file) saved
- When program closed and reopened, **opens with same language selection**
- This setting preserved until manually changed

Usage Scenarios:

Scenario 1: Default Language Usage

- Selected Default option and connected
- Closed program
- **On next launch:** Program opens with Default (English) again

Scenario 2: Custom Language File Usage

- Selected language file from User Define option
- Connected and worked
- Closed program
- **On next launch:** Program automatically opens with language file you selected
- Your last selection appears in language selection box

Advantage: No need to select language file again each time. Once selected, program always opens with that language.

18.6 Language File Not Found

Power Cloud MKII automatically switches to safe mode if saved language file cannot be found.

File Not Found Scenarios:

Saved language file may not be found in following situations:

- Language file **deleted**
- File name **changed**
- File location (folder address) **changed**
- File **moved** to different disk/folder
- File **corrupted** or inaccessible

Automatic Switch to Default Mode:

If language file not found:

- Program searches for file when opening
- Detects file cannot be found
- Automatically switches to **Default (English)** mode
- "Default" selected in language selection box
- User can select User Define again and select file if desired

Solutions:

If your language file not found:

Solution 1: Restore File to Correct Location

- Restore language file from backup
- Place in original location

Solution 2: Reselect File

- Select User Define from language selection box
- Find file's **new location** in file selection window
- Select file and continue
- New location saved

Solution 3: Continue with Default

- Temporarily work with Default (English)
- Fix language file later

> **Recommendation:** Keep your language files in program installation folder or fixed location. This way file path doesn't change and always found. Also take regular backups.

19. Theme Settings Page

Theme Settings page allows you to customize visual appearance of Power Cloud MKII interface. By adjusting color scheme and object sizes according to your preferences, you can achieve more comfortable usage experience.

19.1 Opening Theme Settings Page

Page Access:

Step 1: Click Theme Button

- Press **"Theme"** button from main page
- Theme Settings page opens

Step 2: Theme Settings Displayed

- Preset theme buttons
- Color selection palettes
- Object scale control

19.2 Preset Themes

There are **5 preset theme buttons** on Theme Settings page. These buttons automatically adjust all colors to selected theme tones.

Preset Theme Buttons:

1. Default

- Power Cloud MKII's original color scheme
- Balanced and professional appearance
- Optimized for standard usage

2. Dark

- Dark background, light-colored texts
- Reduces eye fatigue
- Ideal for low light environments
- Modern appearance

3. Light

- Light/white background, dark texts
- Better readability in bright environments
- Classic and clean appearance

- Print-friendly

4. Blue

- Professional appearance in blue tones
- Calm and reliable feeling
- Popular for industrial applications
- Eye-friendly colors

5. Gray

- Neutral appearance in gray tones
- Minimal and modern design
- Non-distracting colors
- Data-focused appearance

Using Preset Themes:

Step 1: Click Theme Button

- Press desired theme button (e.g., **Dark**)
- Theme applied instantly

Step 2: Automatic Color Adaptation

- All 5 colors automatically adjusted to selected theme tones
- Interface instantly updated
- Change immediately visible

Step 3: Save

- Selected theme automatically saved
- Settings stored permanently

19.3 Color Settings

After selecting preset theme or manually, **5 color parameters** available. Custom color selection can be made with color palette for each parameter.

Color Parameters:

1. Primary Text Color

- For description texts and subtitles
- Color of main content texts
- **Usage Areas:**
  - Descriptions
  - Main menu texts
  - Parameter names
  - Important information

2. Secondary Text Color

- Used for headers and important texts
- Less emphasized texts
- **Usage Areas:**
  - Headers
  - Subheaders
  - Helper information

3. Interactive Color

- Selected elements

- **Usage Areas:**
  - Buttons
  - Links
  - Checkboxes
  - Active states
  - Highlight color

4. Background Color

- Main background color
- Ground color of windows and pages
- **Usage Areas:**
  - Main screen background
  - Panel backgrounds
  - Dialog boxes
  - General ground

5. Shadow Color

- Used for shadow effects
- Creates sense of depth
- **Usage Areas:**
  - Button shadows
  - Panel shadows
  - Drop-down menu shadows

Manual Color Selection:

Step 1: Click Color Palette

- Click **color box** next to relevant color parameter
- Color picker opens

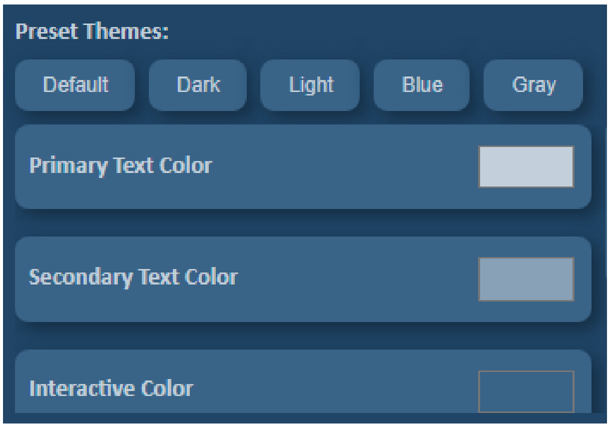
Step 2: Select Color

- Select desired color from color palette
- Can manually enter with RGB, HEX, or HSL values
- **Example:**
  - HEX: #3182CE
  - RGB: rgb(49, 130, 206)
  - HSL: hsl(207, 73%, 57%)

Step 3: Application

- Automatically applied after selecting color
- Interface instantly updated
- Can see change live

Color Detail View:



19.4 Object Scale

Object Scale feature allows you to proportionally change **size of all objects** in interface.

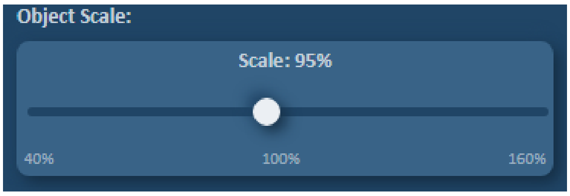
Object Scale Features:

Scale Range:

- **Minimum:** 40%
- **Maximum:** 160%
- **Default:** 100%
- **Increment:** 5% steps

Control Type:

- Adjustment with **Scroll Bar**
- Easy use with drag-drop
- Precise adjustment with value indicator



Using Object Scale:

Step 1: Slide Scroll Bar

- Find **scroll bar** under Object Scale heading
- **Drag** scroll handle left or right
- Left: Reduce (down to 40%)
- Right: Enlarge (up to 160%)

Step 2: Instant Effect

- **All objects instantly scaled** as scroll bar moves
- Change seen live
- Buttons, texts, graphics, all UI elements change proportionally

Step 3: Find Optimal Value

- Find **most comfortable size** by trying different scale values
- Too small: More information visible, harder to read
- Too large: Less information visible, easier to read

Step 4: Auto-Save

- Scale value you selected **automatically saved**
- Setting saved when you release scroll bar

What Gets Scaled:

Object Scale affects all interface elements:

- Text sizes
- Button sizes
- Icon sizes
- Entry boxes
- Tables and lists
- Menus
- Pop-up windows
- All UI elements

Important: Object Scale only affects Power Cloud MKII interface. Doesn't change size of operating system or other programs.

19.5 Saving Settings

All changes made in Theme Settings **automatically saved**.

19.6 Best Practices

Theme Selection:

- Select theme suitable for your work environment
- Bright environment → Light theme
- Dark environment → Dark theme
- Long-term use → Dark or Blue (for eye fatigue)
- If will print → Light theme

Color Harmony:

- Start with preset themes
- Provide high contrast (for text readability)
- Select Interactive Color prominently (button findability)
- Avoid too bright colors
- Consider color blindness

Object Scale:

- Adjust according to your screen size
- Small screen → 80-90%
- Normal screen → 100%
- Large screen → 110-130%
- Presentation/projection → 140-160%
- Vision difficulty → 130-160%

Performance:

- Very large scale may affect performance
- If complex graphics, prefer 100-120% range
- Use 100% on low-performance systems

Standardization:

- Use common theme in team work
- Same appearance in training materials
- Switch to standard settings before screen sharing
- Consistent theme for documentation

20. Layout Selection

Power Cloud MKII software offers screen design in completely customizable structure. With layout (screen arrangement) system that can be edited in JSON file format, you can create numerous custom screen designs according to different usage scenarios and needs, and easily switch between them.

20.1 Layout System Overview

What is Layout?

Layout (Screen Arrangement) is customizable screen design that determines which data will be displayed where and how on Power Cloud MKII interface.

Layout Features:

- **JSON Format:** Layout files stored and editable in JSON format
- **Multiple Layouts:** Multiple different layouts can be created for each device
- **Customizable:** Completely customizable according to user needs
- **Dynamic Switch:** Can switch between layouts while device connection continues
- **Device-Specific:** Separate layouts can be designed for each device model

Layout Usage Areas:

- **Different Operator Profiles:** Different screens for beginner and expert users
- **Application-Based Views:** Special screens for maintenance mode, production mode, test mode
- **Regional Differences:** Custom designs according to different region or customer requirements
- **Data Density:** Simple summary view or detailed analysis view
- **Workflow Optimization:** Screens optimized for specific tasks
- **Screen Size:** Layouts compatible with different screen sizes

20.2 Connecting with Default Layout

Initial Connection:

When you start Power Cloud MKII and connect to device, **default** layout automatically opens.

Default Layout Features:

- Pre-defined standard screen arrangement for each device
- Contains all basic parameters and controls
- Optimized for general usage

Connection Steps:

- Open Power Cloud MKII
- Press **Connect** button
- Connection established to device
- **Default layout** automatically loads and displays on screen

20.3 Selecting Different Layout

You can switch to different layout while connection continues.

Layout Change Steps:

Step 1: Click Layout Button

- While connected to device, press **"Layout"** button
- Layout selection window opens

Important: To press Layout button, you must first be connected to device. Layout selection cannot be made without connection.

Step 2: Select Desired Layout

- Click layout you want to use from list
- Press **"Open"** button
- Selected layout loads instantly
- Screen rearranged according to new layout
- **Connection continues without interruption**

Step 3: Work with New Layout

- Now selected layout active
- Screen design changed
- Different data displays and controls may appear
- Can continue working normally

20.4 Layout Change Features

Seamless Transition:

- **Device connection doesn't drop** during layout change
- Data flow continues
- No parameter or setting lost
- Instant transition provided

Dynamic Refresh:

- Screen automatically rearranged according to new layout
- Invisible parameters hidden
- Parameters defined in new layout become visible

Advantage: By quickly switching between different layouts for different tasks, you can optimize your workflow. For example, you can use simple layout for routine monitoring, detailed layout for troubleshooting.

20.5 Multiple Layout Management

Unlimited Layouts:

- **Many layout files** can be created for each device
- No theoretical limit on number of layouts
- Can prepare dozens of different views according to your needs

Layout Organization:

Organize by Categories:

- **User Level:** Operator, Technician, Engineer
- **Operation Type:** Monitoring, Maintenance, Test
- **Project:** Project A screen, Project B screen

Layout Naming Recommendations:

- **Use Descriptive Names:**
    - Operator_Simple_View.json
    - Engineer_Advanced_Monitoring.json
    - Maintenance_Service_Panel.json
- **Category Prefixes:**
    - OP_ (Operator)
    - ENG_ (Engineer)
    - MAINT_ (Maintenance)
    - PROD_ (Production)

- **Version Information:**
  - Dashboard_v1.0.json
  - Dashboard_v2.1_updated.json

21. Documents Page

Documents page allows you to access technical documentation, user manuals, and other reference materials related to connected device. With this page, you can easily download and use device documents.

21.1 Opening Documents Page

Button Location and Visibility:

Documents Button:

- Located in **lower right corner** of main page
- Becomes **visible only after connection established to device**
- This button not visible before connection established

Page Access:

Step 1: Connect to Device

- Establish connection to device with Connect button
- Complete login process
- Main screen opens

Step 2: Click Documents Button

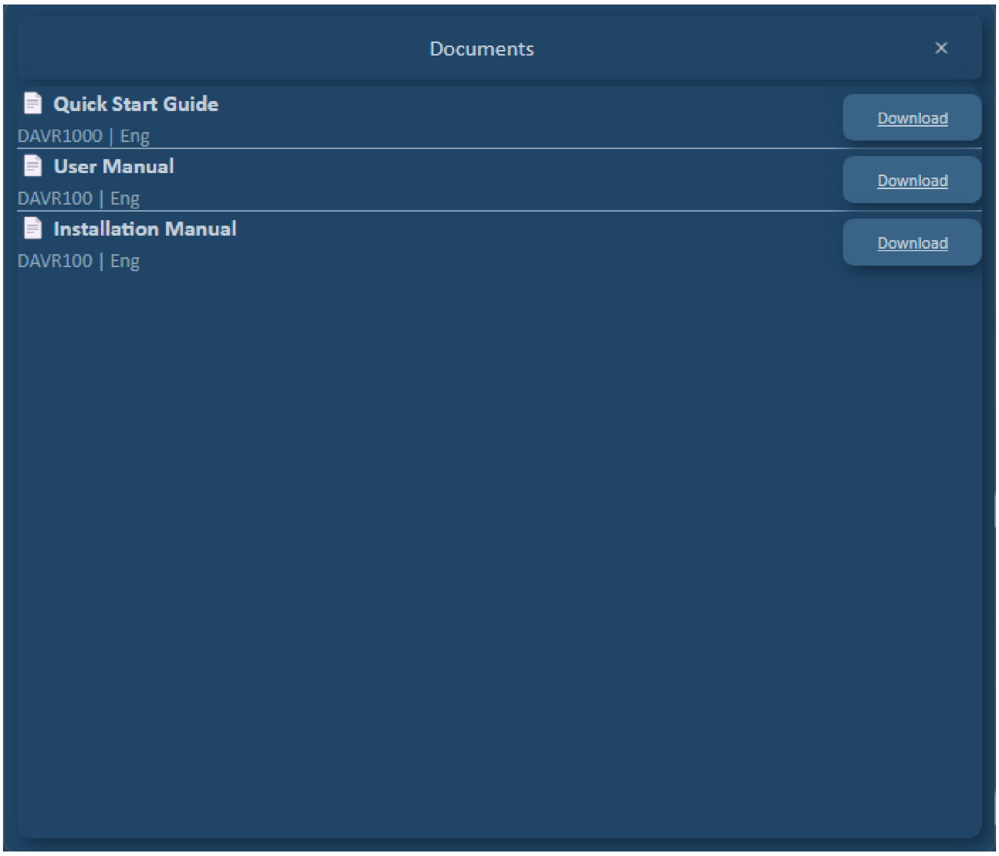
- Press **"Documents"** button in lower right corner
- Documents page opens

Step 3: Document List Loads

- Internet connection checked
- Documents related to device pulled from internet
- List displayed on screen

21.2 Documents Page Structure

Page View:



21.3 Document List Fields

Document list consists of four fields:

1. Document Name

- Document name and version information
- Descriptive and identifying name
- **Examples:**
  - "User Manual v2.3"
  - "Quick Start Guide"
  - "Technical Datasheet"
  - "Installation Manual"
  - "Wiring Diagram"
  - "Maintenance Guide"
  - "Software Update Instructions"

2. Device Name

- Which device document is for
- Model name of connected device
- All documents will be for same device

3. Document Language

- Which language document is written in

4. Action - Download Button

- **"Download"** button at end of each row
- Used to download document
- Clickable

21.4 Document Download

Separate Download button for each document.

Download Process Steps:

Step 1: Find Desired Document

- View document you want to download in list
- Check document name, language

Step 2: Click Download Button

- Press **"Download"** button on relevant row
- Download process starts

Step 3: File Save Window

- Your browser's or operating system's file save window opens
- Select **save location**
- **File name** comes automatically (you can change if desired)

Step 4: Download

- Press **Save** button
- File downloaded to location you specified

Step 5: Download Complete

- File ready in folder you specified
- You can open and use file

21.5 Internet Connection Requirement

Important: Documents page **requires internet connection**.

Internet Connection Available:

- Document list successfully loads
- All documents related to device visible
- Download buttons active and usable
- Documents downloaded from server

No Internet Connection:

- Document list comes **empty**
- No documents shown
- Warning message displayed (Document not found)

21.6 Document Types and Usage Areas

Document types that may be found on Documents page:

User Manual:

- General usage instructions for device
- Detailed explanation of all features
- Step-by-step user guide
- **Usage:** Initial setup and daily operation

Quick Start Guide:

- Basic setup steps
- Initial startup instructions
- Short and concise information
- **Usage:** Quick start and basic setup

Technical Datasheet:

- Technical features and specifications
- Electrical parameters
- Dimensions and physical properties
- **Usage:** Technical planning and integration

Installation Manual:

- Physical installation steps
- Assembly instructions
- Connection diagrams
- **Usage:** Initial setup and assembly

Software Update Instructions:

- Firmware update steps
- Update notes
- **Usage:** Software updates

21.7 Troubleshooting

Problem 1: "Documents Button Not Visible"

- **Solution:**
  - Ensure connected to device
  - Complete login process
  - Check button in lower right corner

Problem 2: "Document List Empty"

- **Solution:**
  - Check your internet connection
  - Check your proxy settings
  - Try again on different network

Problem 3: "Download Not Working"

- **Solution:**
  - Check internet connection

Problem 4: "Downloaded File Won't Open"

- **Solution:**

- Ensure PDF reader installed
- Check if download completed
- File may be corrupt, download again
- Try different PDF reader
- Check file format

Problem 5: "Got Document in Wrong Language"

- **Solution:**
  - Check Document Language column
  - Ensure selected document in correct language
  - Contact manufacturer if document not available in desired language